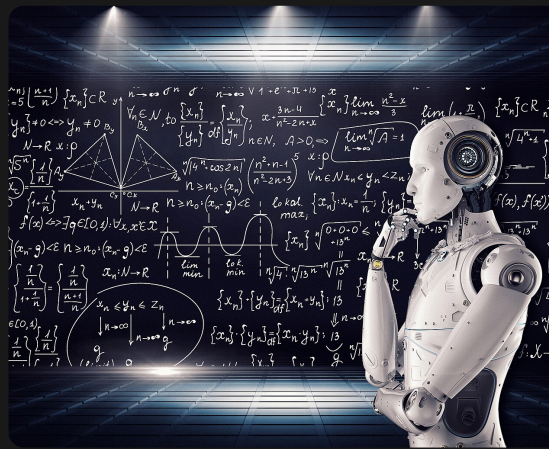




EUROPEAN
SPALLATION
SOURCE

Andrew Sazonov
ESS DMSC



WHAT AI CAN CHANGE IN CRYSTALLOGRAPHIC COMPUTING

Early lessons from building
a diffraction engine from scratch

Andrew Sazonov
ESS DMSC

OUTLINE

💡 Motivation

🏗️ What was built

🕒 Speed and correctness

🔧 How it was built

OUTLINE

💡 Motivation

🏗️ What was built

🕒 Speed and correctness

🔧 How it was built

AI IN SCIENTIFIC SOFTWARE

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

AI/ML named in the [title or abstract](#)

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

AI/ML named in the [title or abstract](#)

23

of 142 microsymposia

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

AI/ML named in the **title or abstract**

23

of 142 microsymposia

121

of 839 microsymposium talks

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

AI/ML named in the **title or abstract**

23

of 142 microsymposia

121

of 839 microsymposium talks

69

of 932 posters

HOW AI ENTERS SOFTWARE DEVELOPMENT

HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX

Is it fast?

1.5 - 41×

faster than the old parser

same Python data
structures,
single keyword to switch

1.3 - 1.5×

faster than Gemmi

native path, matched
output;
5-27% less peak memory

8.1 s

for a 3.3 GB file

file sizes that were
previously
impossible become routine

Benchmarks as a decision instrument:

- Does the added code complexity pay for itself in speed?
- Does an optimisation help everywhere, or only on typical files?

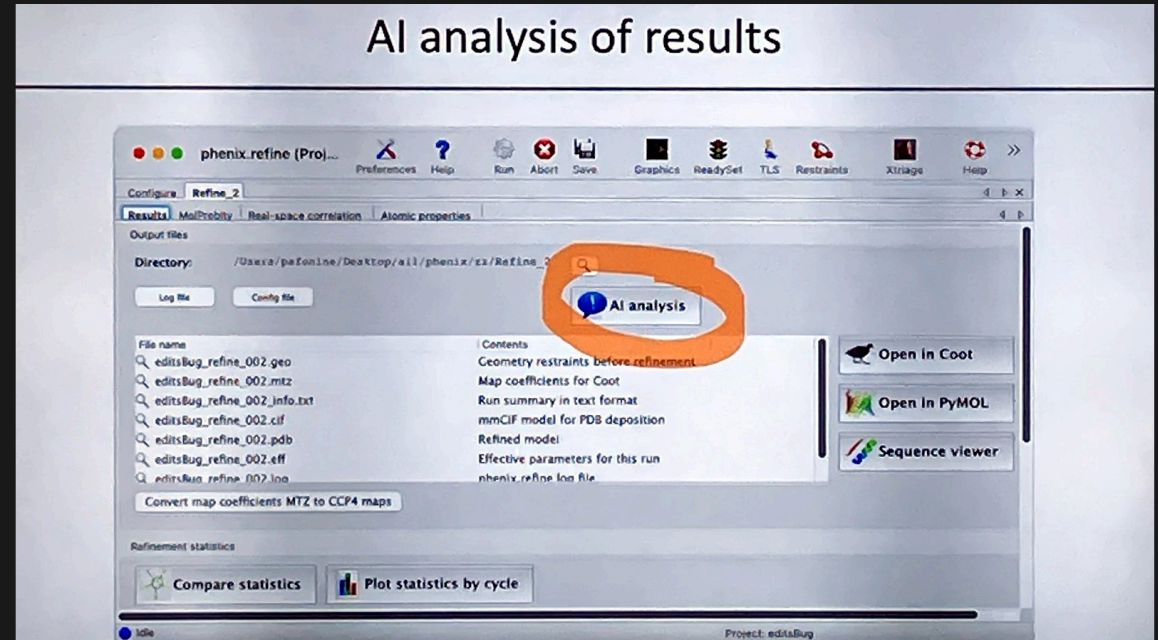
HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX

② AI assistants and chatbots

mature product · Phenix



HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX

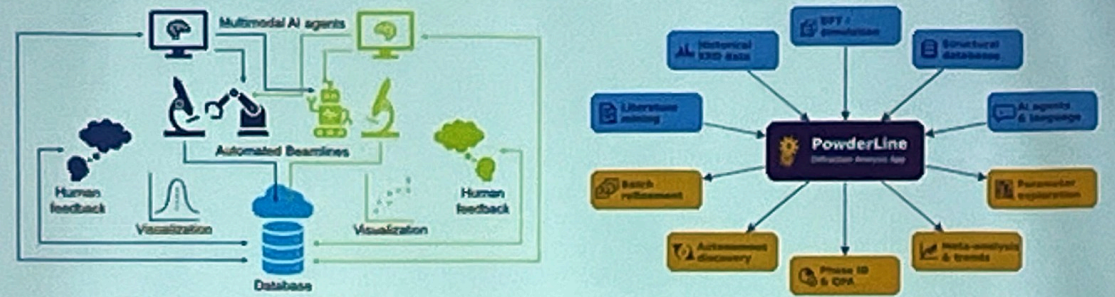
② AI assistants and chatbots

mature product · Phenix

③ AI orchestration

decides what to measure next, then processes it · NSLS-II

Summary and looking forward



- At NSLS-II we have developed a modular framework that enables collaborative human-AI operating modes on Bluesky-enabled instruments
- PowderLine transforms highly manual, pathway-dependent diffraction analyses into automatable, scalable, and reproducible tasks
- Together, these tools and AI-enhanced workflows enable a paradigm shift toward hypothesis driven autonomous experiments and accelerated discovery

HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX

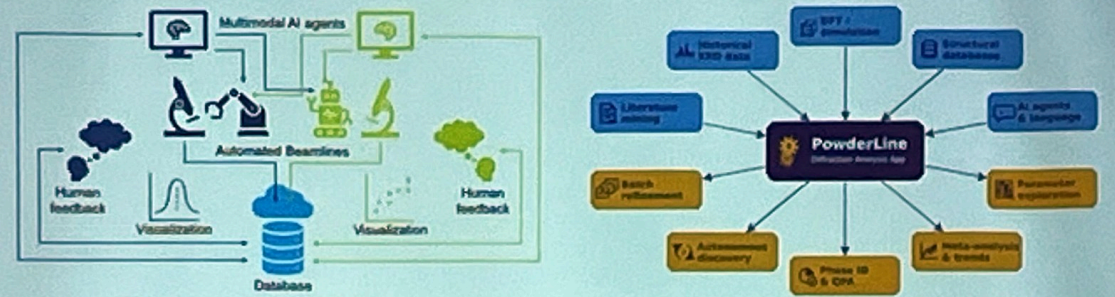
② AI assistants and chatbots

mature product · Phenix

③ AI orchestration

decides what to measure next, then processes it · NSLS-II

Summary and looking forward



- At NSLS-II we have developed a modular framework that enables collaborative human-AI operating modes on Bluesky-enabled instruments
- PowderLine transforms highly manual, pathway-dependent diffraction analyses into automatable, scalable, and reproducible tasks
- Together, these tools and AI-enhanced workflows enable a paradigm shift toward hypothesis driven autonomous experiments and accelerated discovery

This talk focuses on ①

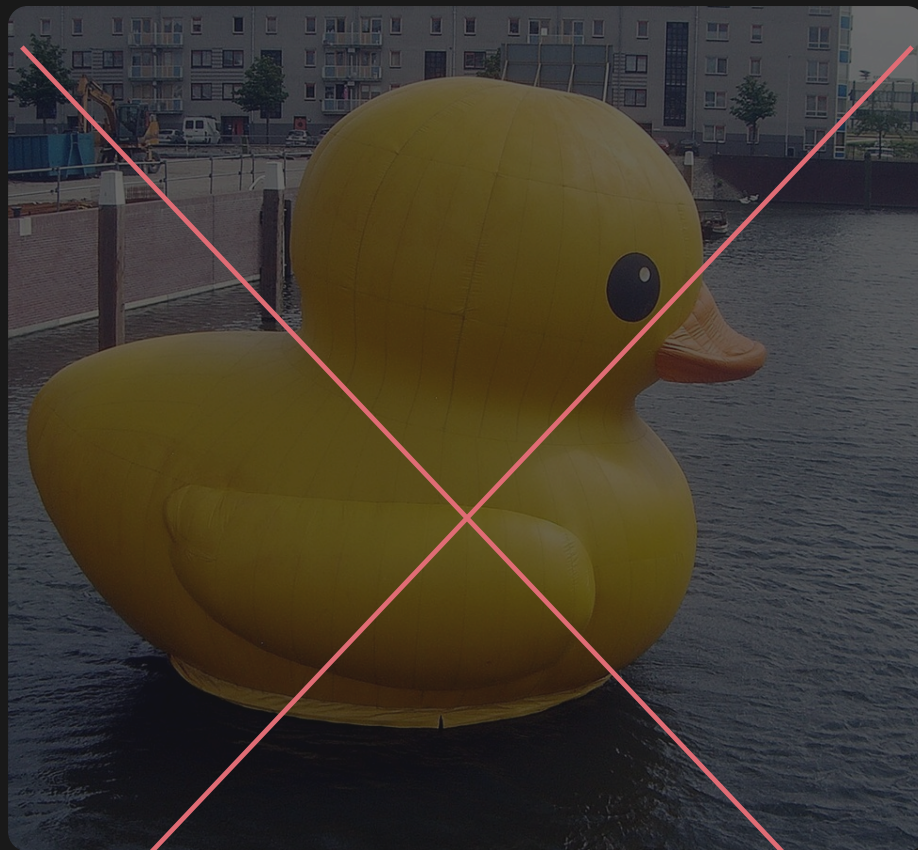
WHY?

WHY?



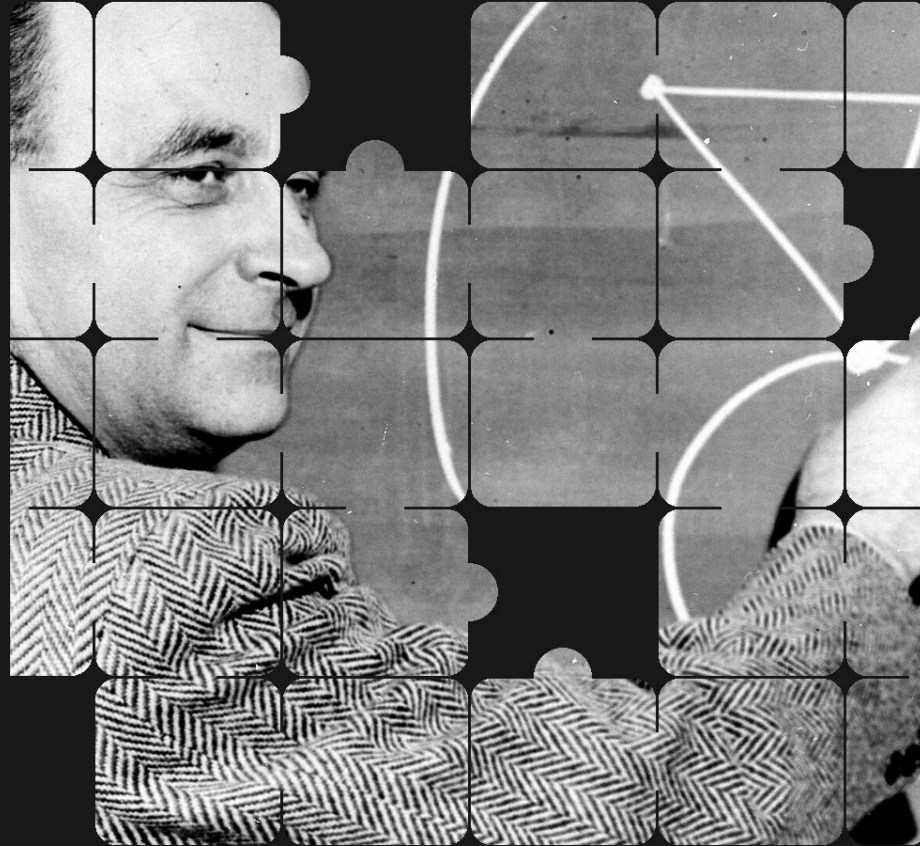
Just for fun

WHY?



Just for fun

WHY?



To fill gaps in
the existing approach

THE EXISTING APPROACH

THE EXISTING APPROACH

Reuse third-party calculation and fitting engines

THE EXISTING APPROACH

Reuse third-party calculation and fitting engines

Calculation engines

CrysPy
Python

CrysFML
Fortran → Python

PDFfit2
C++ → Python

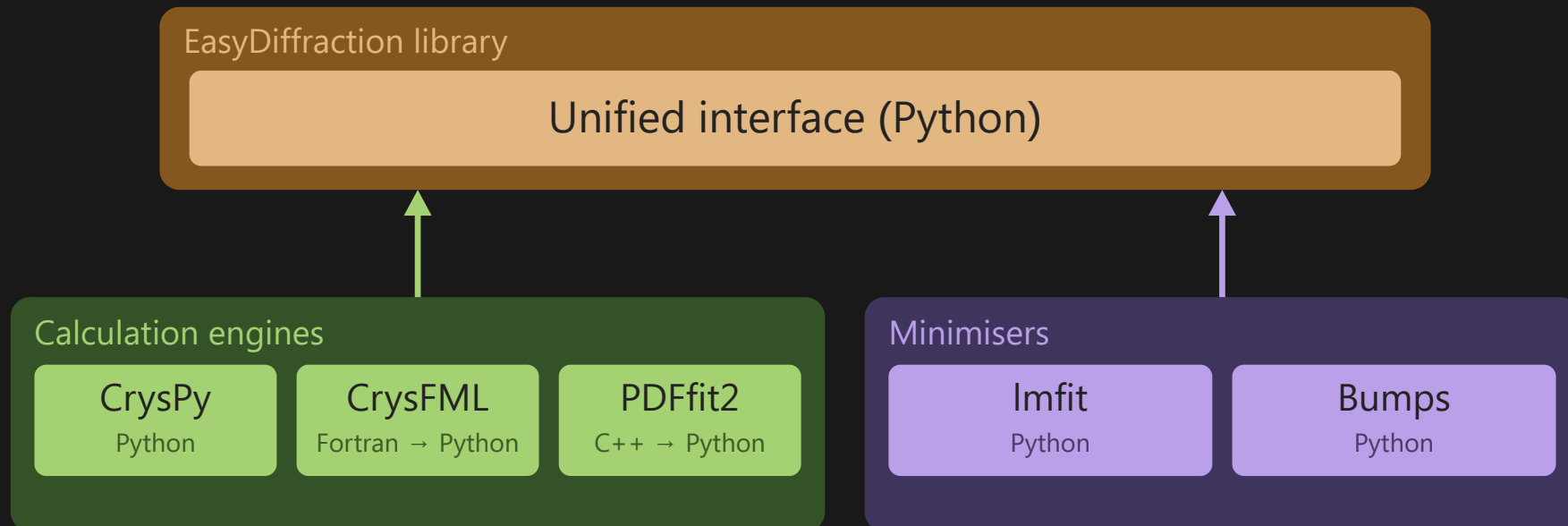
Minimisers

Imfit
Python

Bumps
Python

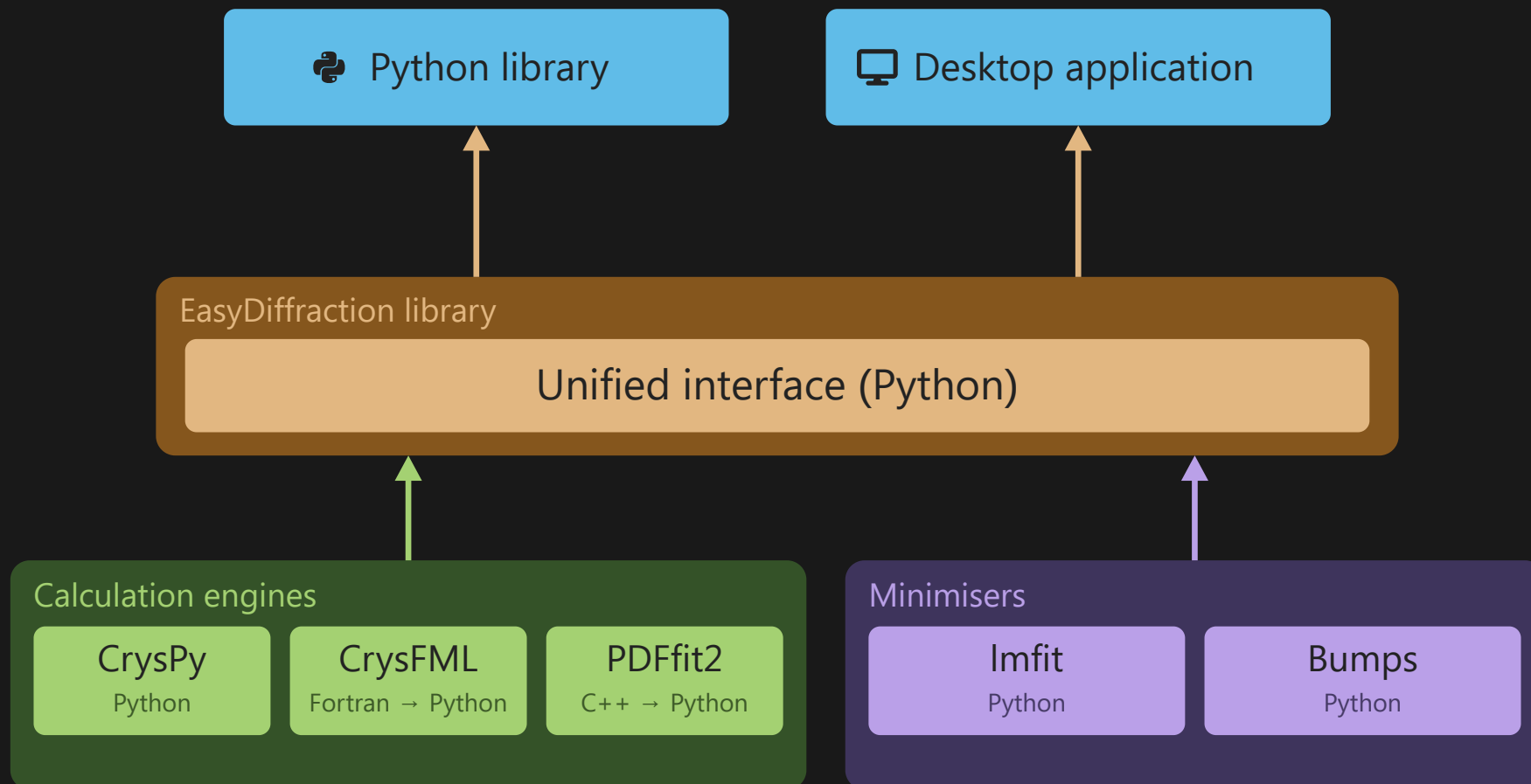
THE EXISTING APPROACH

Reuse third-party calculation and fitting engines



THE EXISTING APPROACH

Reuse third-party calculation and fitting engines



WHAT IT COSTS US

WHAT IT COSTS US



Maintenance

wrappers and bindings for many libraries

WHAT IT COSTS US



Maintenance

wrappers and bindings for many libraries



Performance

overhead, non-optimised fitting

WHAT IT COSTS US



Maintenance

wrappers and bindings for many libraries



Performance

overhead, non-optimised fitting



Extensibility

new features may require changes in several libraries

WHAT MODERN ANALYSIS NEEDS

WHAT MODERN ANALYSIS NEEDS

- 📊 **Large series** · up to 1000 datasets × 100 patterns each
temperature, field and time series from one experiment

WHAT MODERN ANALYSIS NEEDS

 **Large series** · up to 1000 datasets × 100 patterns each
temperature, field and time series from one experiment

 **Multidimensional data** and very large parameter spaces
2D and 3D detectors, joint refinement of many patterns at once

WHAT MODERN ANALYSIS NEEDS

 **Large series** · up to 1000 datasets × 100 patterns each
temperature, field and time series from one experiment

 **Multidimensional data** and very large parameter spaces
2D and 3D detectors, joint refinement of many patterns at once

 **Automation** · experiments that run themselves
sequencing and recovery without a human · a bad dataset must not stop the run

WHAT MODERN ANALYSIS NEEDS

 **Large series** · up to 1000 datasets × 100 patterns each
temperature, field and time series from one experiment

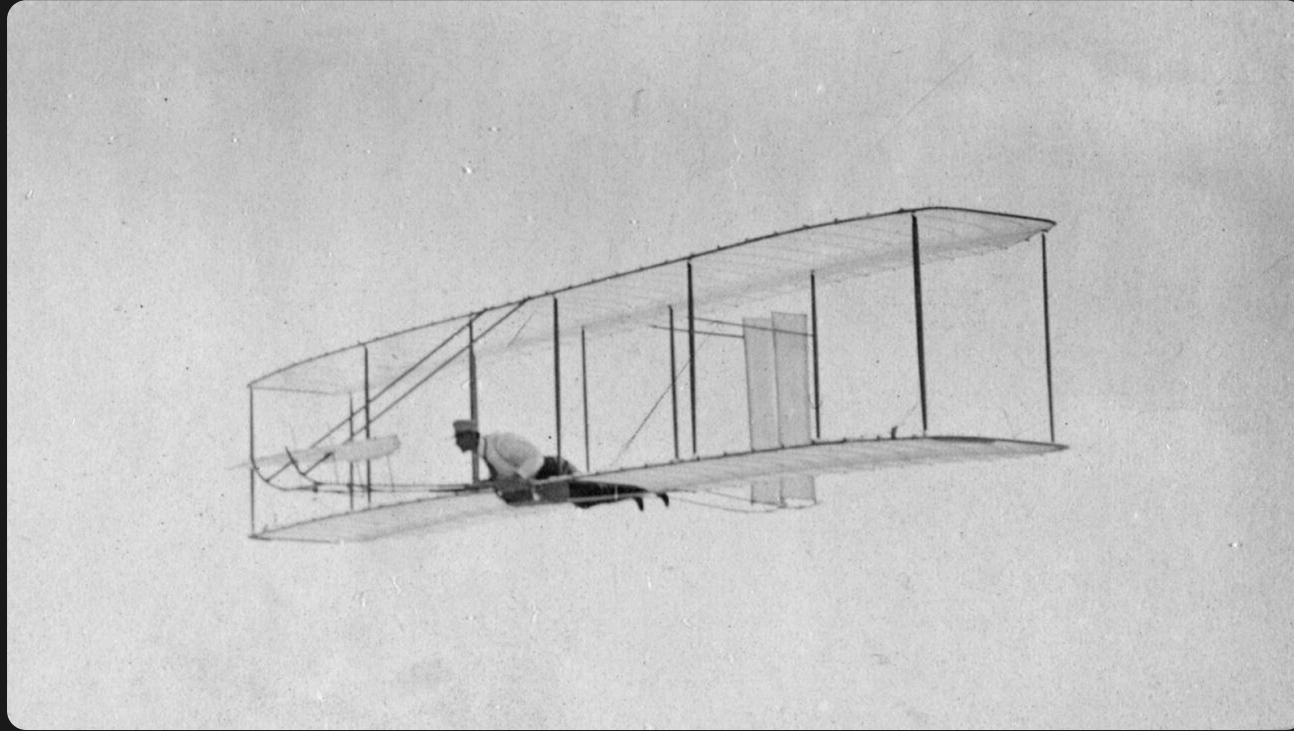
 **Multidimensional data** and very large parameter spaces
2D and 3D detectors, joint refinement of many patterns at once

 **Automation** · experiments that run themselves
sequencing and recovery without a human · a bad dataset must not stop the run

As the data grows, **performance matters more and more**

THE EXPERIMENT

THE EXPERIMENT



How far can we get if **AI writes every line**
of a real crystallographic engine?

OUTLINE

💡 Motivation

🏗️ What was built

🕒 Speed and correctness

🔧 How it was built

ONE ENGINE, FOUR INTERFACES

ONE ENGINE, FOUR INTERFACES

crysta

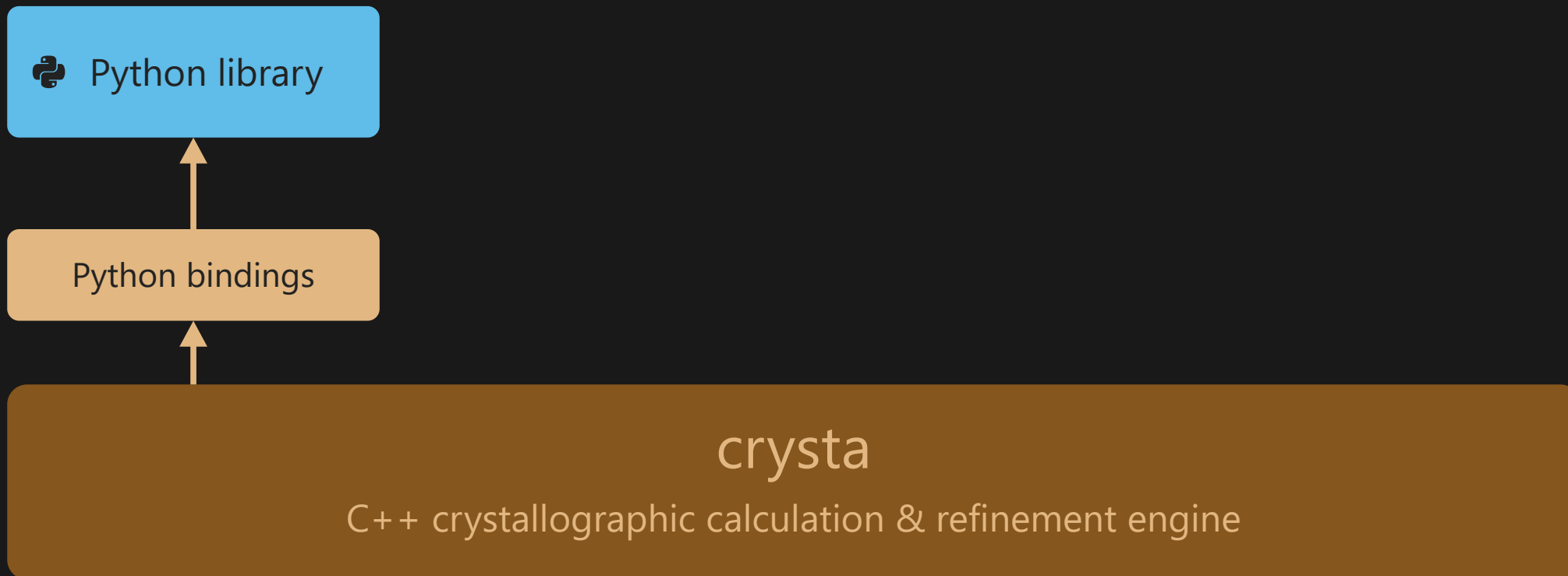
ONE ENGINE, FOUR INTERFACES

crysta

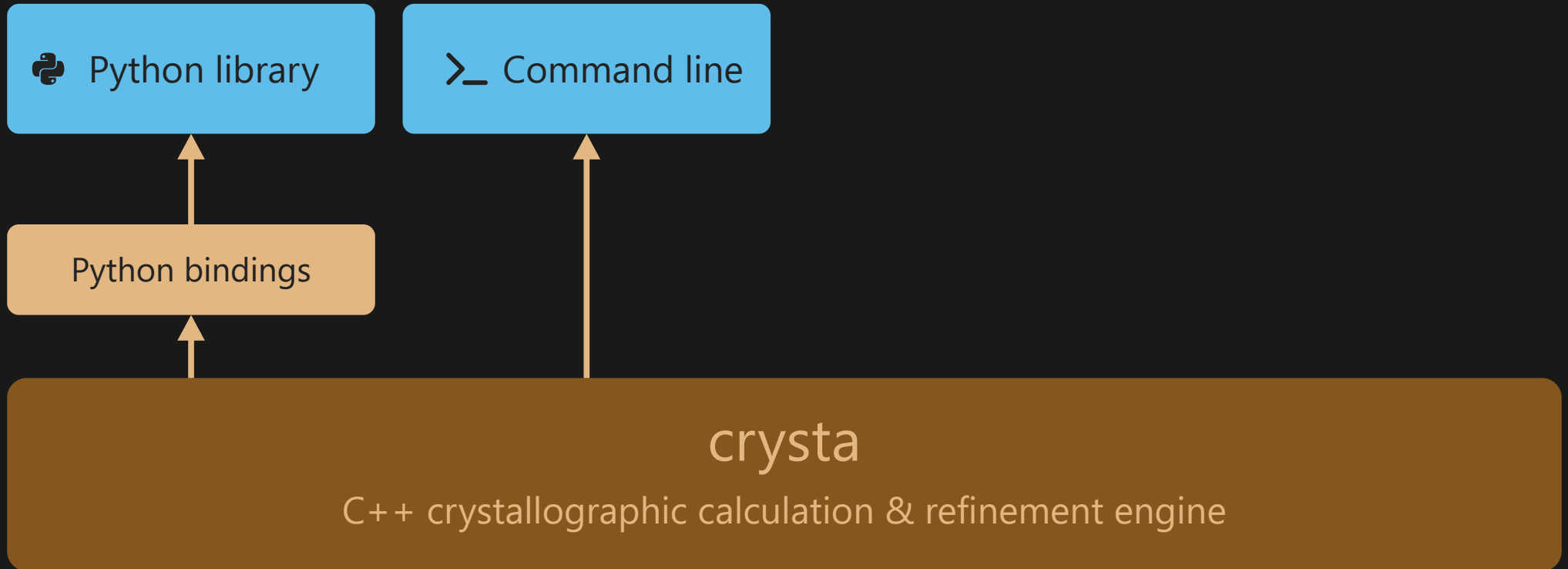
C++ crystallographic calculation & refinement engine



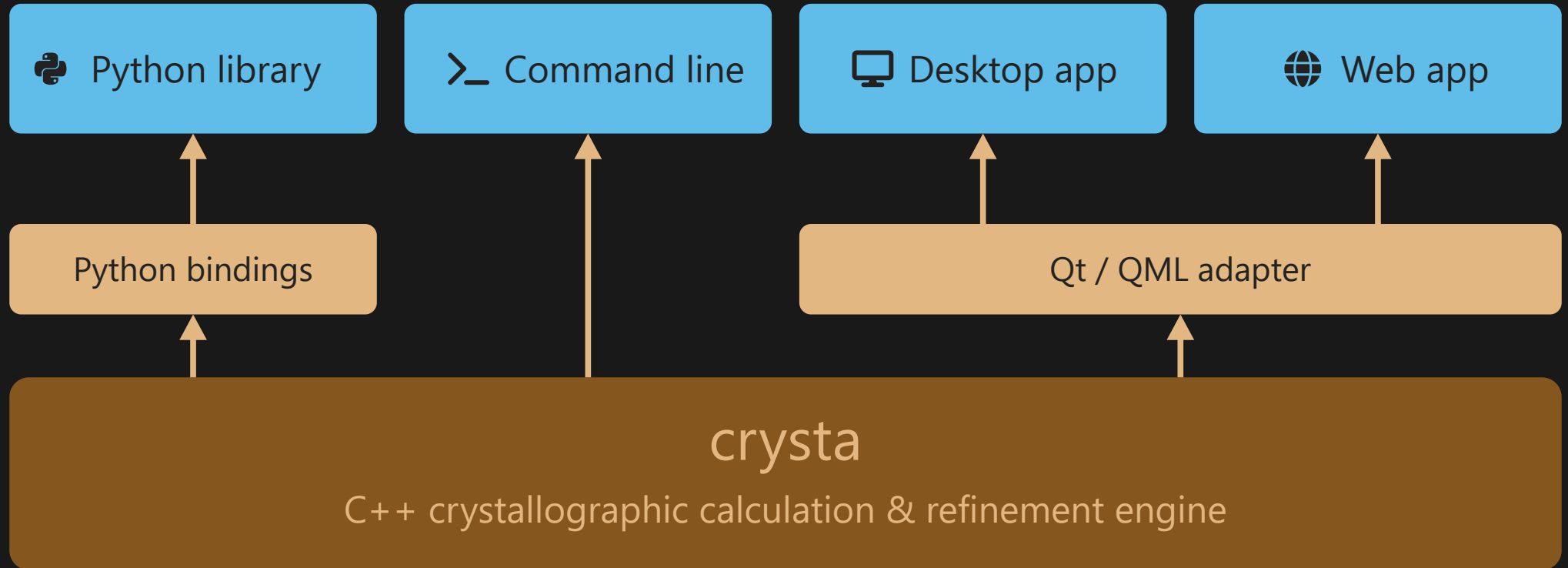
ONE ENGINE, FOUR INTERFACES



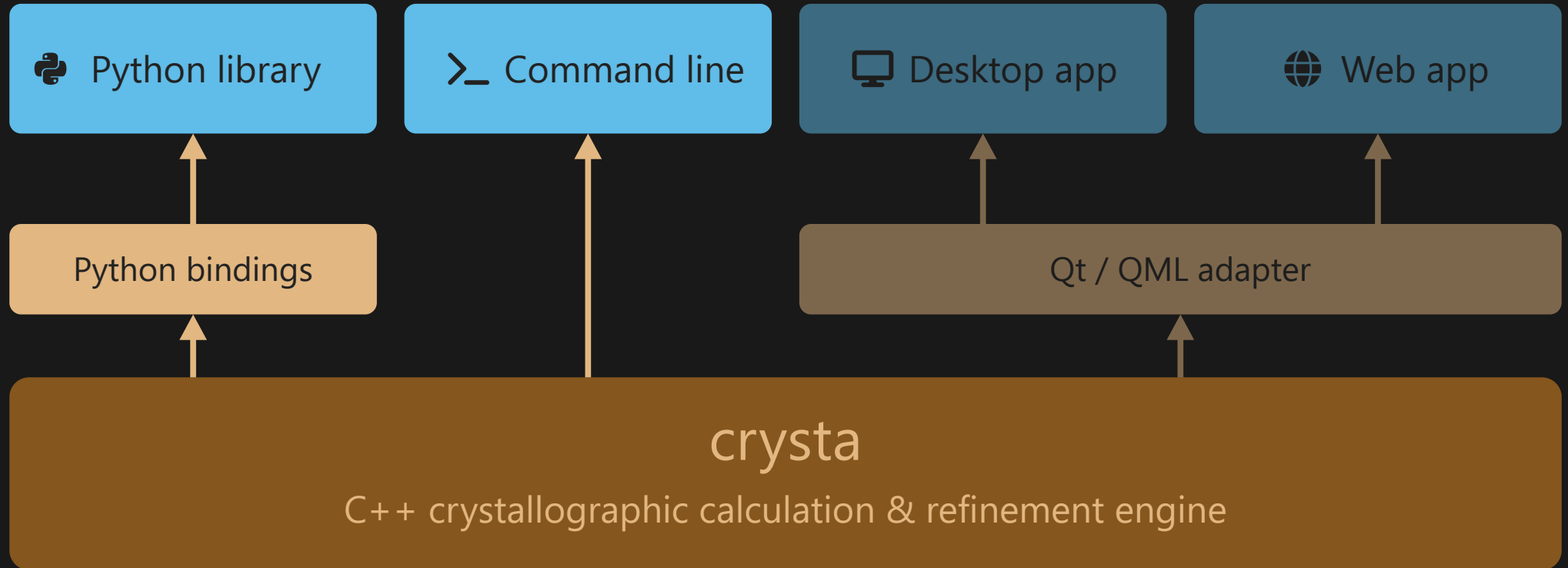
ONE ENGINE, FOUR INTERFACES



ONE ENGINE, FOUR INTERFACES



ONE ENGINE, FOUR INTERFACES



WHAT IT STANDS ON

WHAT IT STANDS ON

In the product

Eigen

linear algebra · the only one required

OpenMP

threads · optional — else it runs serially

SLEEFPiece

vector maths · optional — else one at a time

nanobind

the Python bindings · native builds only

WHAT IT STANDS ON

In the product

Eigen

linear algebra · the only one required

OpenMP

threads · optional — else it runs serially

SLEEF

vector maths · optional — else one at a time

nanobind

the Python bindings · native builds only

In development

CMake · Ninja · scikit-build-core

configure, build and package the wheels

pixi

the environment, pinned for everyone

doctest · pytest

the tests, C++ and Python

ruff · clang-tidy · cppcheck · mypy

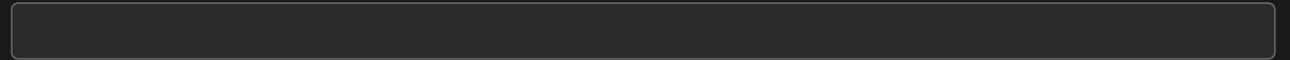
static analysis

HOW MUCH IS DONE

HOW MUCH IS DONE

relative coverage from feature lists and my estimates

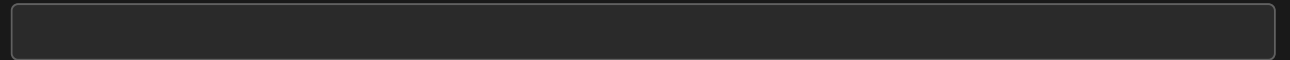
Structure model



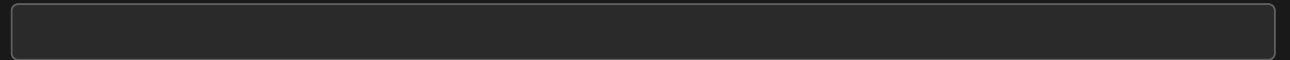
Peak profiles



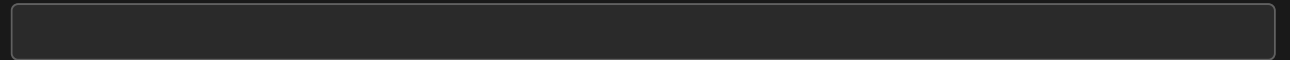
Instrument & corrections



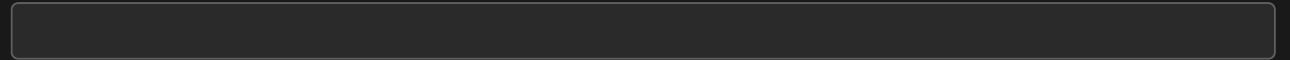
Background, cutoff, texture



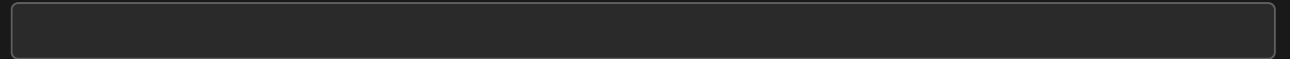
Le Bail, pair distribution



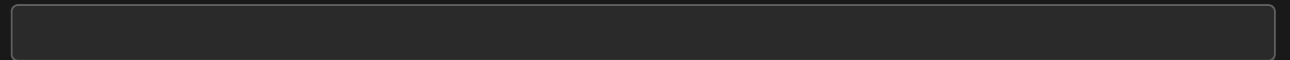
Magnetic structures



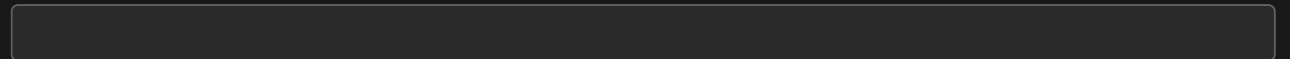
Single crystal



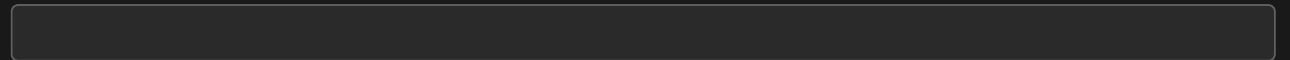
Calculation modes, multi-dataset



Minimisers & sampling



Fit strategies & constraints

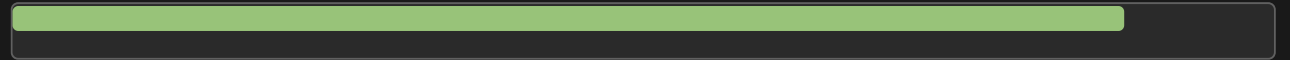


HOW MUCH IS DONE

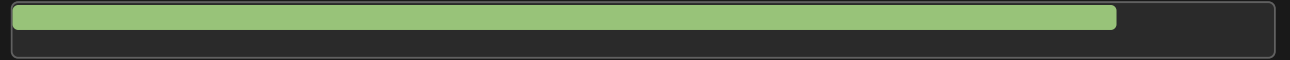
relative coverage from feature lists and my estimates

EasyDiffraction today

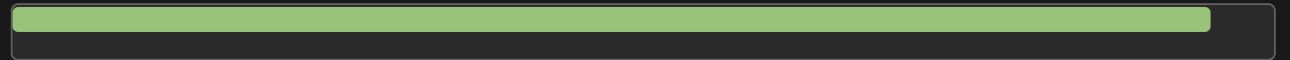
Structure model



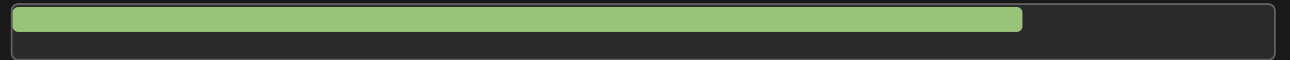
Peak profiles



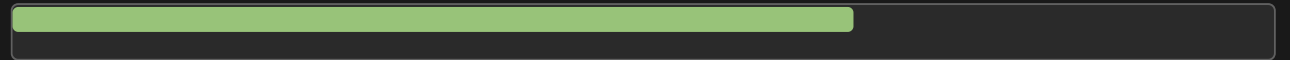
Instrument & corrections



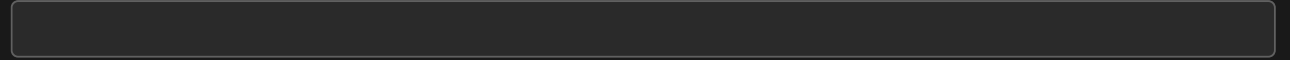
Background, cutoff, texture



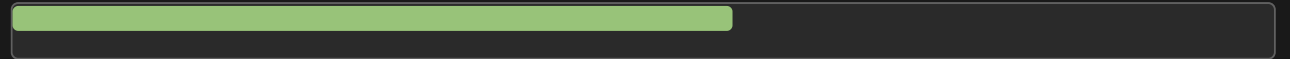
Le Bail, pair distribution



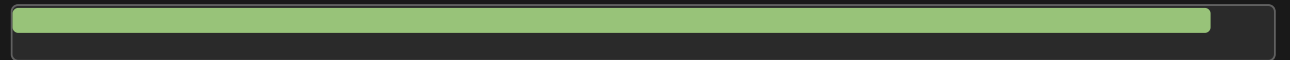
Magnetic structures



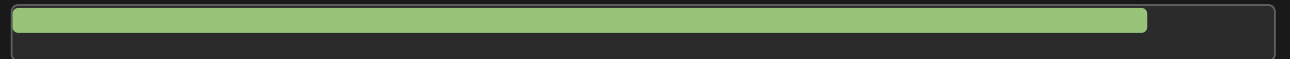
Single crystal



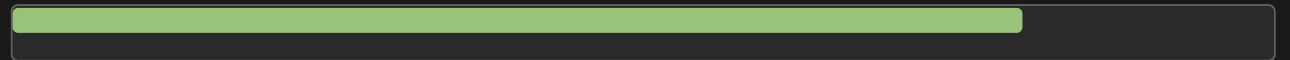
Calculation modes, multi-dataset



Minimisers & sampling



Fit strategies & constraints



HOW MUCH IS DONE

relative coverage from feature lists and my estimates

EasyDiffraction today

crysta today

Structure model

Peak profiles

Instrument & corrections

Background, cutoff, texture

Le Bail, pair distribution

Magnetic structures

Single crystal

Calculation modes, multi-dataset

Minimisers & sampling

Fit strategies & constraints

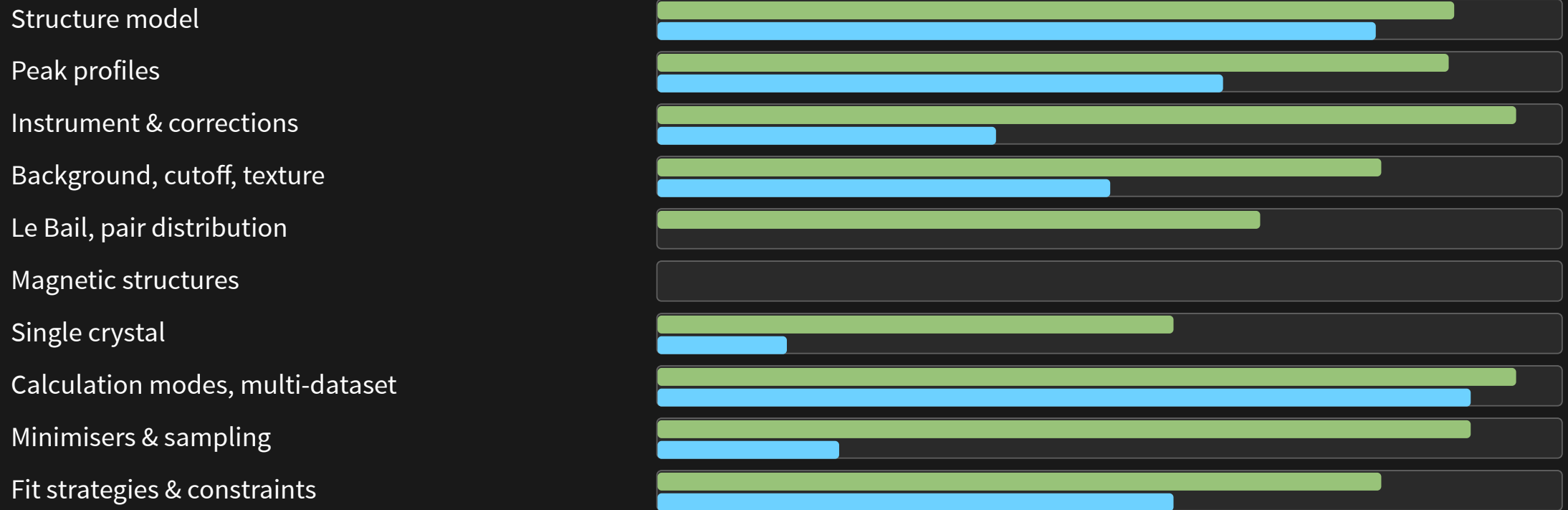


HOW MUCH IS DONE

relative coverage from feature lists and my estimates

EasyDiffraction today

crysta today



About **eight weeks** of development

OUTLINE

💡 Motivation

🏗️ What was built

🕒 Speed and correctness

🔧 How it was built

THE TEST CASE

THE TEST CASE

 Sample · NCAF, $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$

a standard reference powder · cubic $I2_13$, $a = 10.2503 \text{ \AA}$, six sites

THE TEST CASE

⚙ Sample · NCAF, $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$

a standard reference powder · cubic $I2_13$, $a = 10.2503 \text{ \AA}$, six sites

📐 Measurement · WISH at ISIS, banks 2 + 9 merged

one pattern at $2\theta = 58.33^\circ$ · 3977 points, 627 peaks in the fitted range

THE TEST CASE

⚙ **Sample** · NCAF, $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$

a standard reference powder · cubic $I2_13$, $a = 10.2503 \text{ \AA}$, six sites

📐 **Measurement** · WISH at ISIS, banks 2 + 9 merged

one pattern at $2\theta = 58.33^\circ$ · 3977 points, 627 peaks in the fitted range

⚙ **Refinement** · 48 free parameters

coordinates, displacement, scale, zero, profile, background · same start for every engine

THE TEST CASE

⚙️ **Sample** · NCAF, $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$

a standard reference powder · cubic $I2_13$, $a = 10.2503 \text{ \AA}$, six sites

📐 **Measurement** · WISH at ISIS, banks 2 + 9 merged

one pattern at $2\theta = 58.33^\circ$ · 3977 points, 627 peaks in the fitted range

⚙️ **Refinement** · 48 free parameters

coordinates, displacement, scale, zero, profile, background · same start for every engine

⚖️ **Reference** · FullProf CLI 8.40

fp2k · same model and instrument parameters · every difference is measured against it

THE TEST CASE

⚙️ **Sample** · NCAF, $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$

a standard reference powder · cubic $I2_13$, $a = 10.2503 \text{ \AA}$, six sites

📐 **Measurement** · WISH at ISIS, banks 2 + 9 merged

one pattern at $2\theta = 58.33^\circ$ · 3977 points, 627 peaks in the fitted range

⚙️ **Refinement** · 48 free parameters

coordinates, displacement, scale, zero, profile, background · same start for every engine

⚖️ **Reference** · FullProf CLI 8.40

fp2k · same model and instrument parameters · every difference is measured against it

🔧 **Machine** · VISA, an Intel Xeon node · 6 cores

runs use 1 or 6 threads, as labelled

SPEED: ONE PATTERN CALCULATION

 NCAF

 3977 points

 627 peaks

 WISH 2+9

 VISA 6 cores

 log scale

SPEED: ONE PATTERN CALCULATION

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📡 WISH 2+9

🖨️ VISA 6 cores

📊 log scale

Python prototype (no NumPy)

1 ms

10 ms

100 ms

1 s

10 s



4.49 s

SPEED: ONE PATTERN CALCULATION

🧪 NCAF

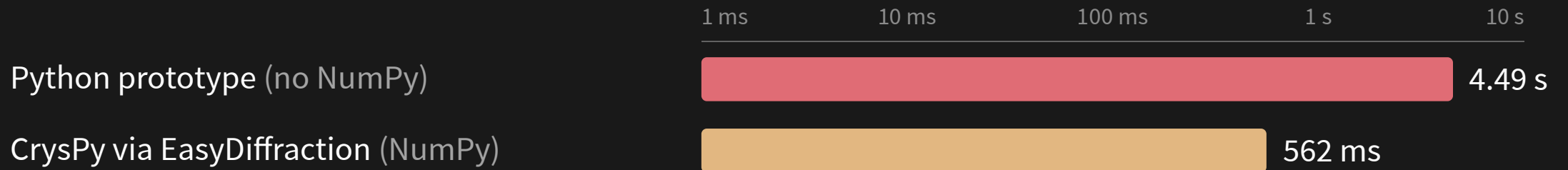
📈 3977 points

⬆️ 627 peaks

📡 WISH 2+9

⚙️ VISA 6 cores

📊 log scale



SPEED: ONE PATTERN CALCULATION

🧪 NCAF

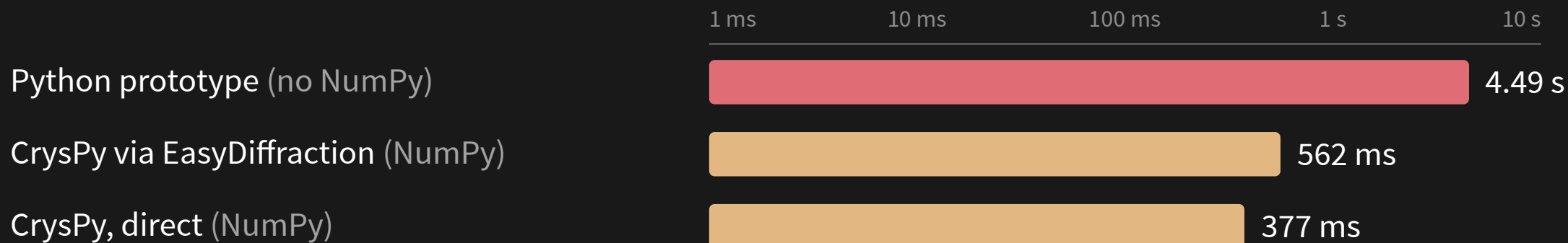
📈 3977 points

⬆️ 627 peaks

🔌 WISH 2+9

🖨️ VISA 6 cores

📊 log scale



SPEED: ONE PATTERN CALCULATION

🧪 NCAF

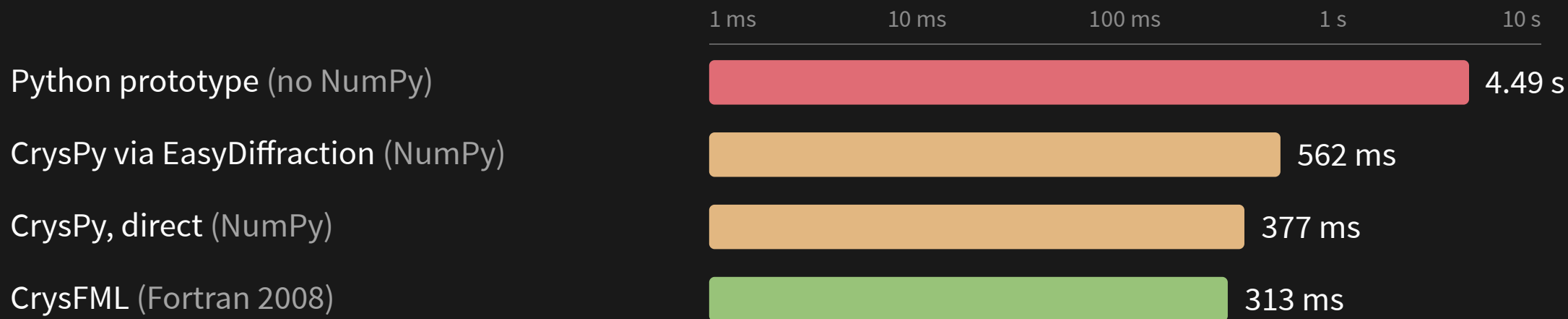
📈 3977 points

⬆️ 627 peaks

🔧 WISH 2+9

🖨️ VISA 6 cores

📊 log scale



SPEED: ONE PATTERN CALCULATION

🧪 NCAF

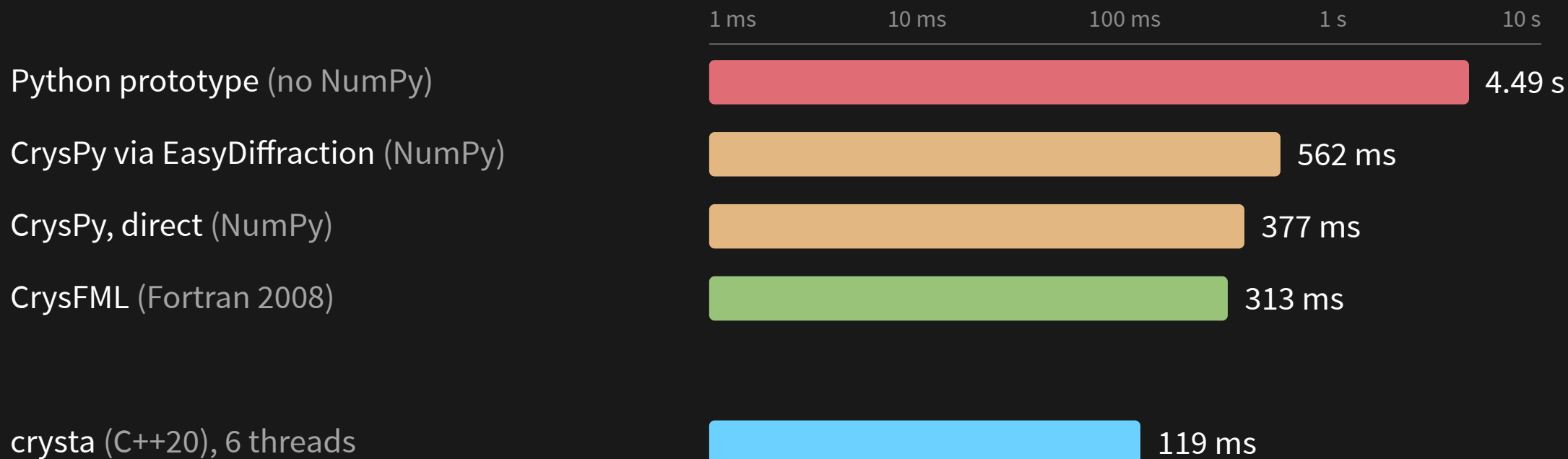
📈 3977 points

⬆️ 627 peaks

🔧 WISH 2+9

🖨️ VISA 6 cores

📊 log scale



SPEED: ONE PATTERN CALCULATION

🧪 NCAF

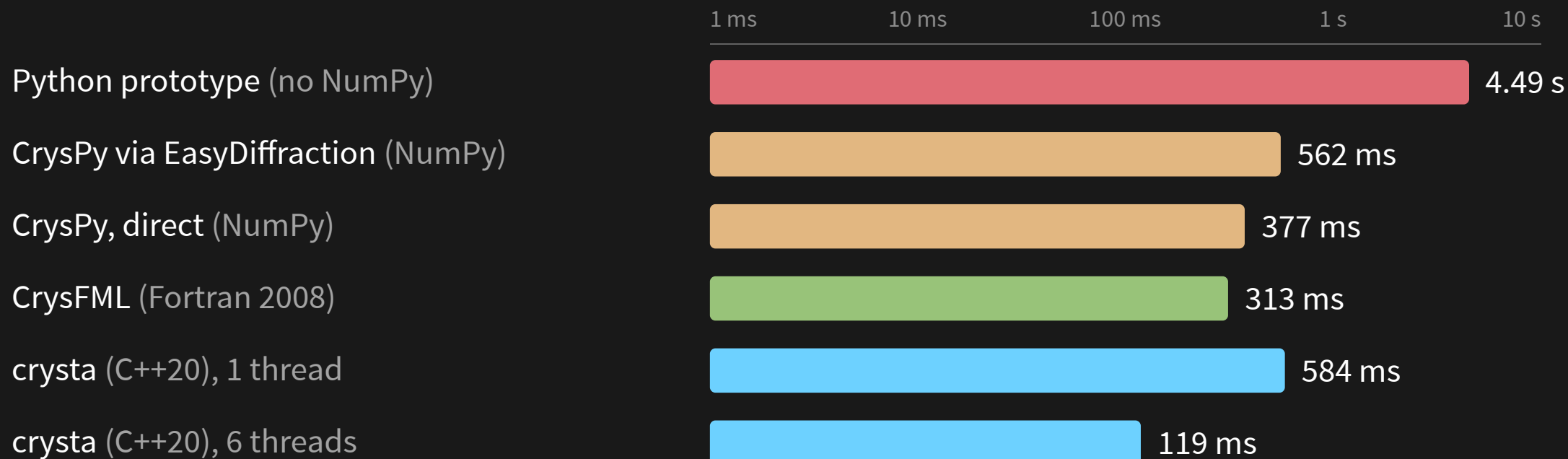
📈 3977 points

⬆️ 627 peaks

🔧 WISH 2+9

⚙️ VISA 6 cores

📊 log scale



SPEED: ONE PATTERN CALCULATION

🧪 NCAF

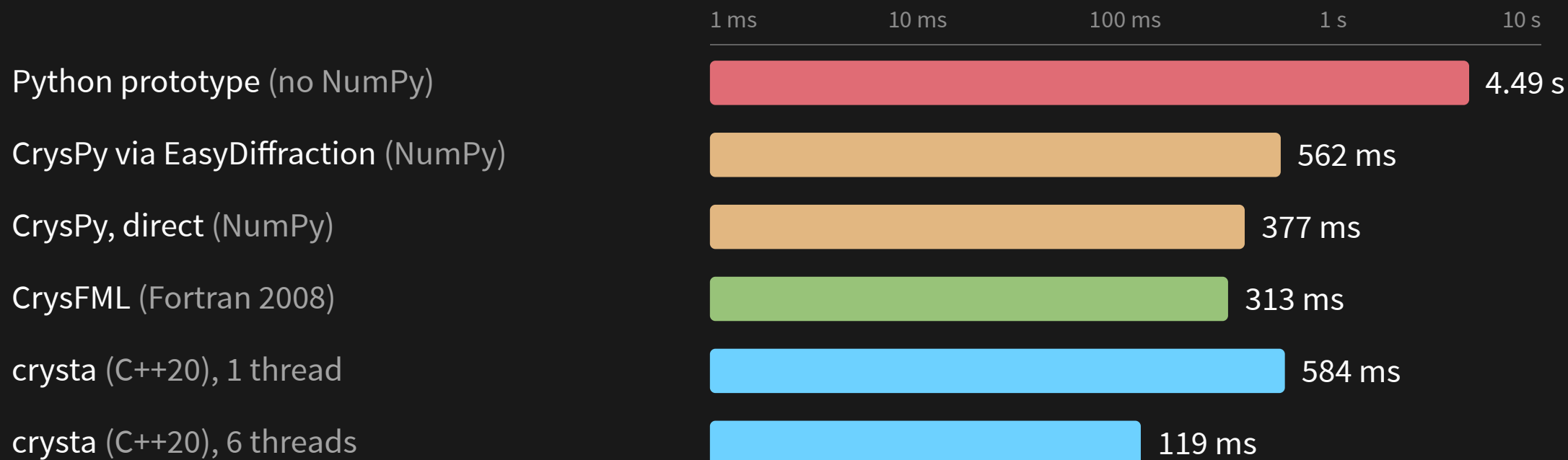
📈 3977 points

⬆️ 627 peaks

🔧 WISH 2+9

⚙️ VISA 6 cores

📊 log scale



✂️ Recomputing reflections that symmetry makes identical

SPEED: ONE PATTERN CALCULATION

🧪 NCAF

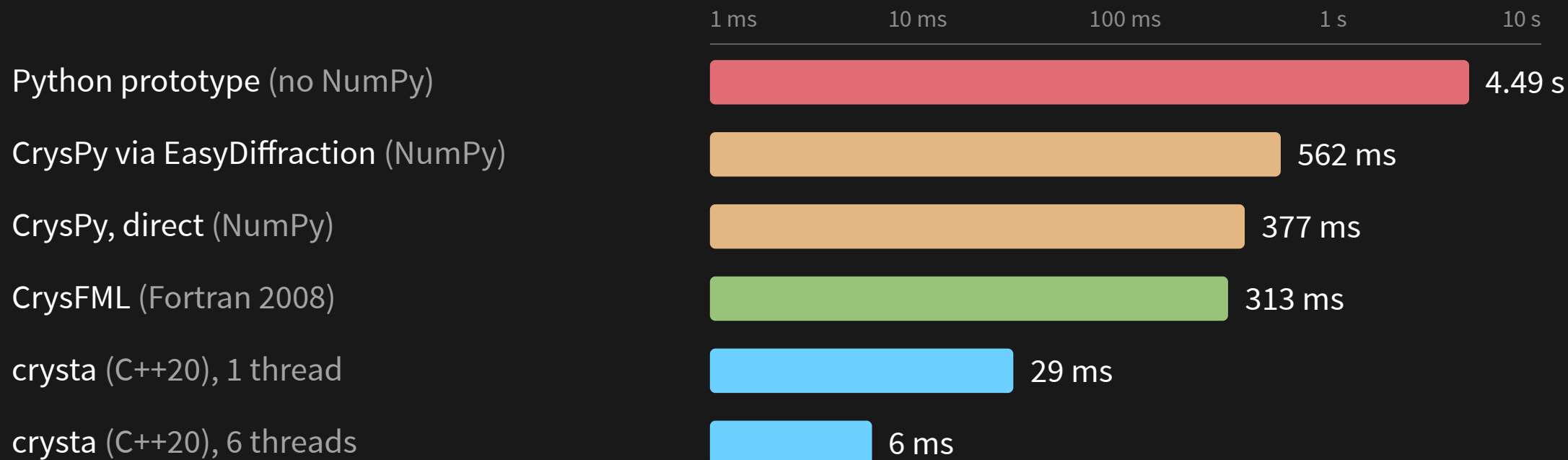
📈 3977 points

⬆️ 627 peaks

🔊 WISH 2+9

⚙️ VISA 6 cores

📊 log scale



✂️ Recomputing reflections that symmetry makes identical

AGREEMENT OF CALCULATED PATTERNS

 NCAF

 3977 points

 627 peaks

 WISH 2+9

 VISA 6 cores

AGREEMENT OF CALCULATED PATTERNS

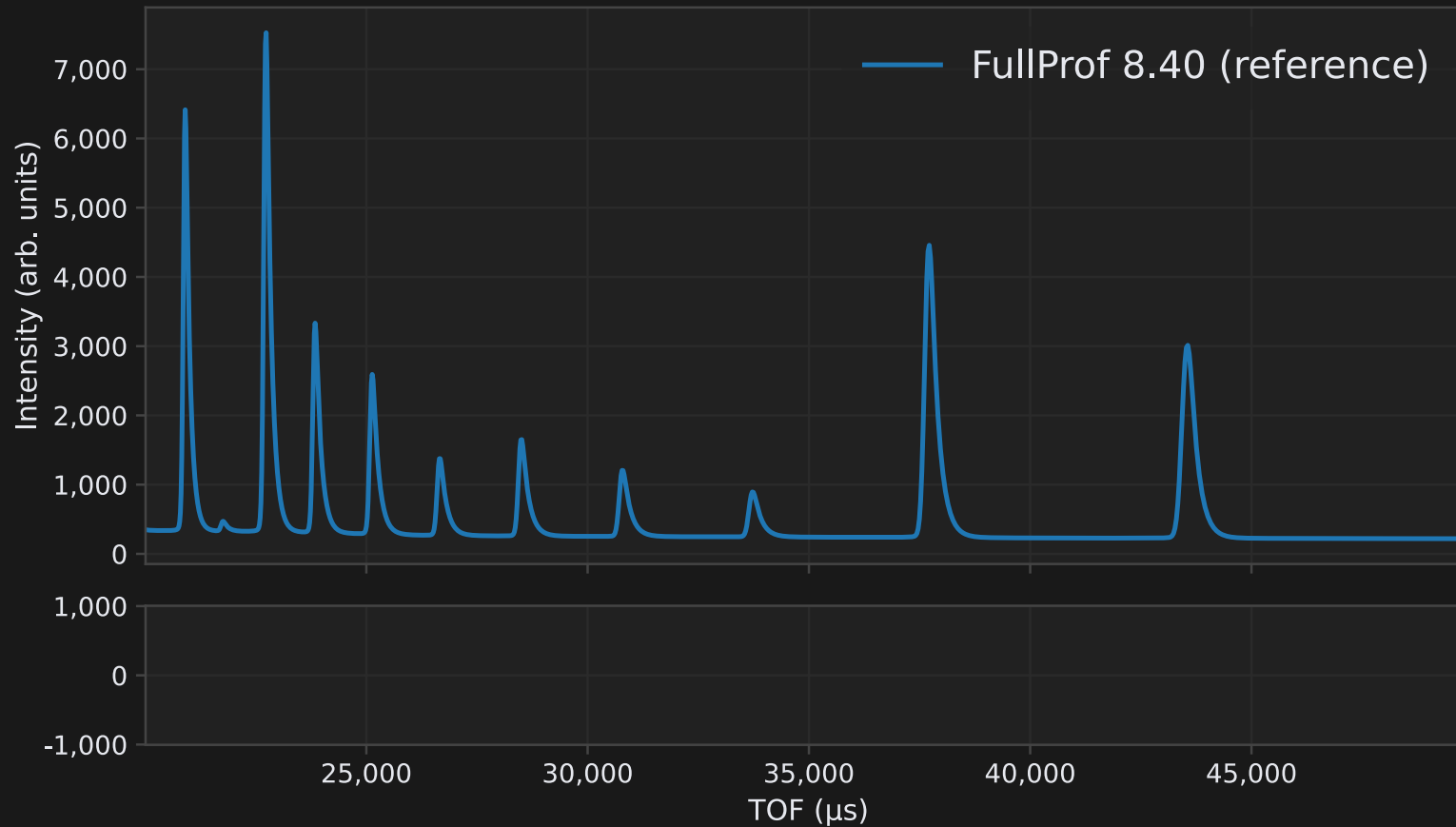
🧪 NCAF

📈 3977 points

📐 627 peaks

📡 WISH 2+9

💻 VISA 6 cores



AGREEMENT OF CALCULATED PATTERNS

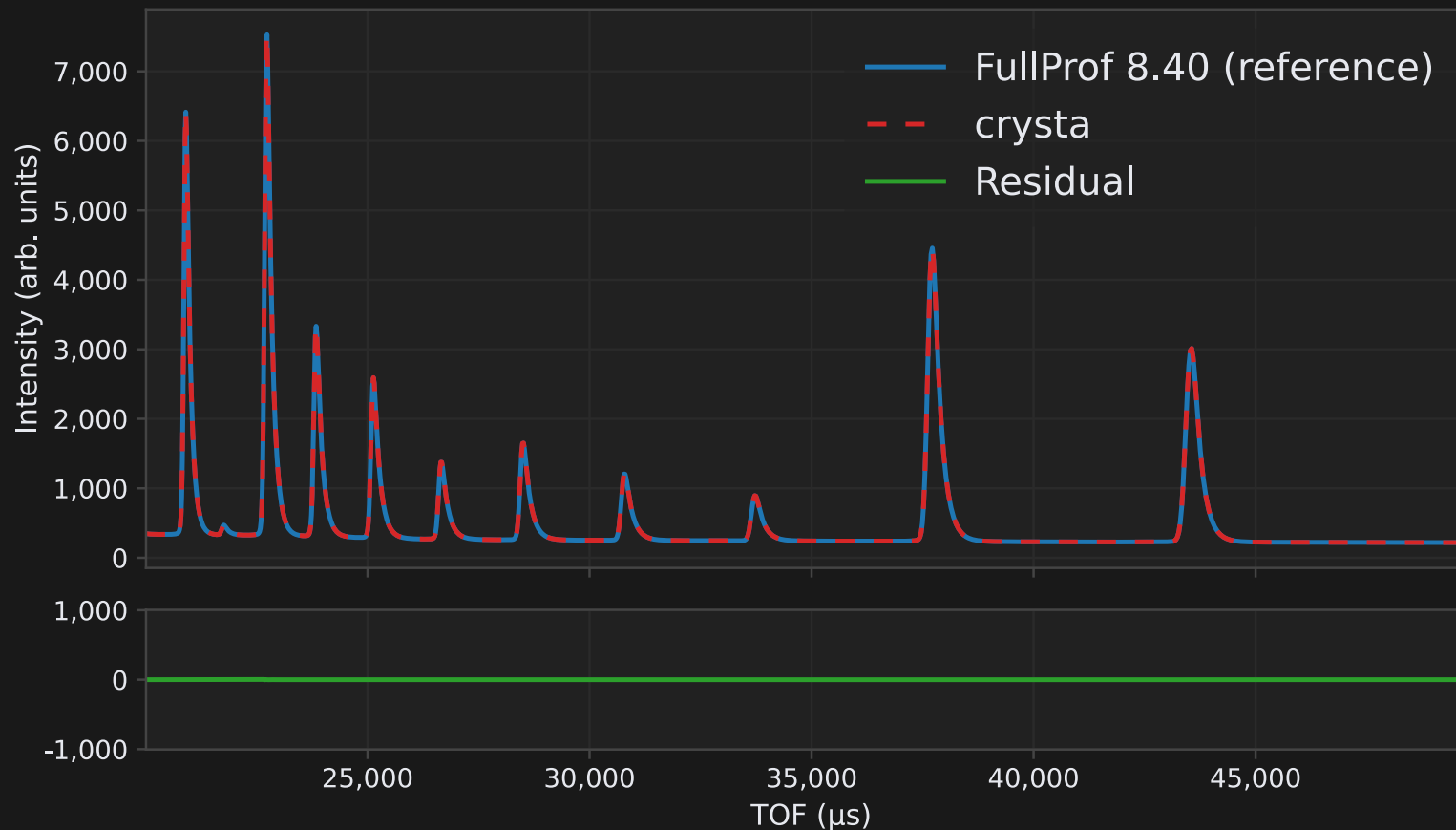
🧪 NCAF

📈 3977 points

📐 627 peaks

📡 WISH 2+9

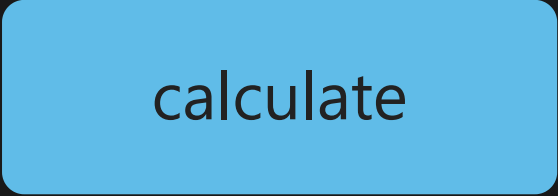
💻 VISA 6 cores



Largest difference 3.9 / 7527 peak counts — 0.05 %

A FIT IS NOT ONE CALCULATION

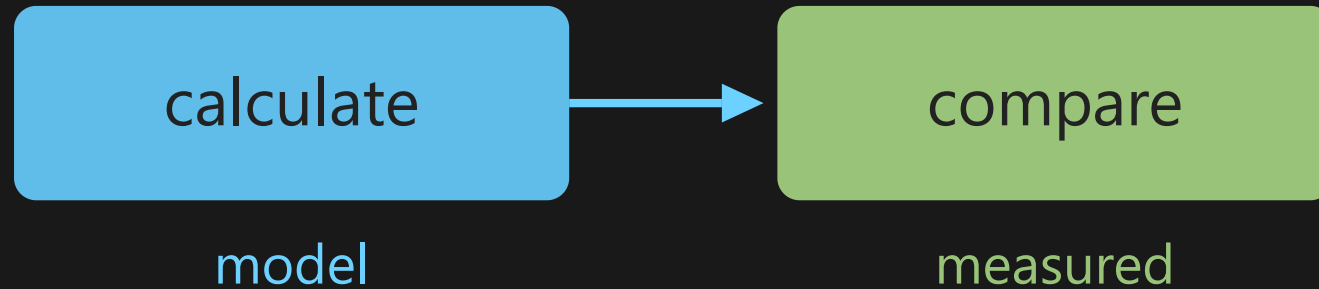
A FIT IS NOT ONE CALCULATION



calculate

model

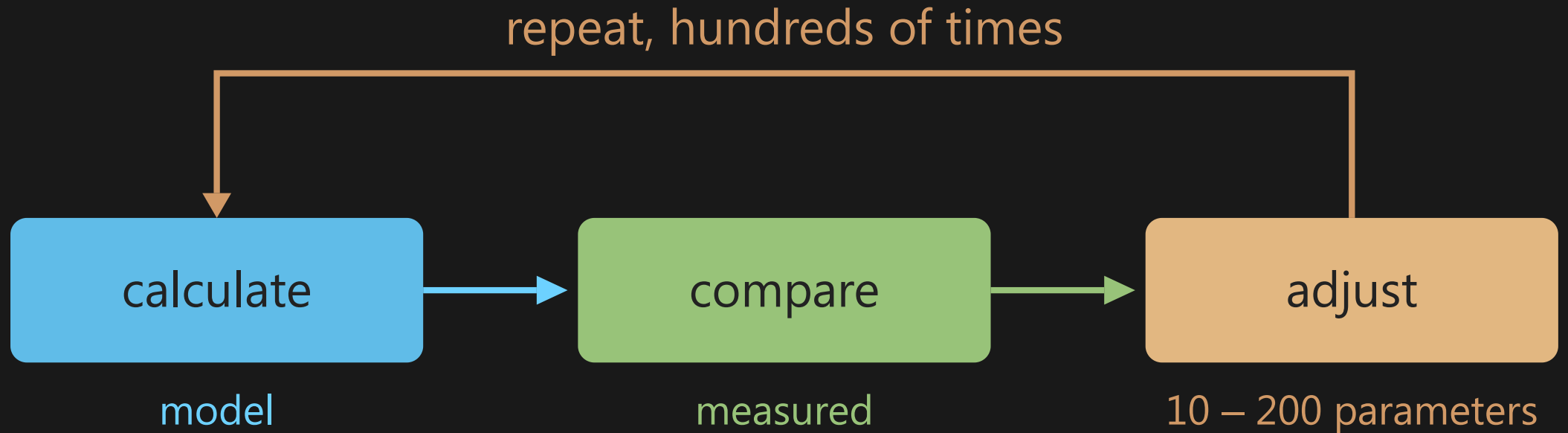
A FIT IS NOT ONE CALCULATION



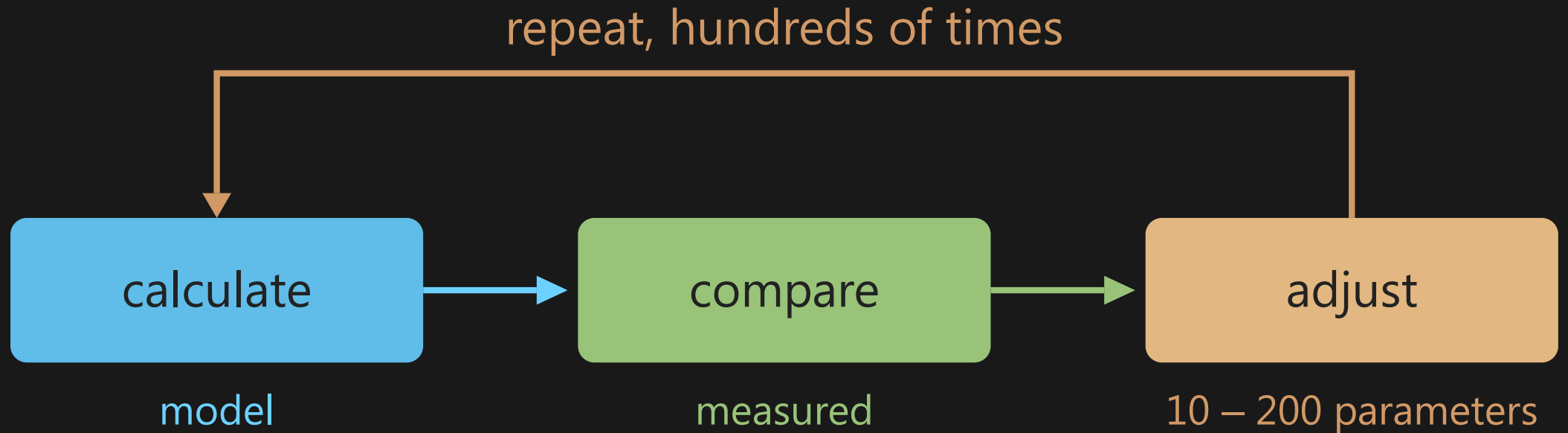
A FIT IS NOT ONE CALCULATION



A FIT IS NOT ONE CALCULATION



A FIT IS NOT ONE CALCULATION



Adjusting needs **derivatives** — how the pattern moves with a parameter

THREE WAYS TO GET DERIVATIVES

THREE WAYS TO GET DERIVATIVES

- ① **Numerical** (EasyDiffraction) · change, recalculate, subtract
finite differences · one extra full pattern per parameter, per step · approximate

THREE WAYS TO GET DERIVATIVES

① **Numerical** (EasyDiffraction) · change, recalculate, subtract

finite differences · one extra full pattern per parameter, per step · approximate

② **Analytic** (FullProf) · derive formulas and write them by hand

every profile × every parameter · exact, but hundreds of formulas to maintain

THREE WAYS TO GET DERIVATIVES

① **Numerical** (EasyDiffraction) · change, recalculate, subtract
finite differences · one extra full pattern per parameter, per step · approximate

② **Analytic** (FullProf) · derive formulas and write them by hand
every profile × every parameter · exact, but hundreds of formulas to maintain

③ **Automatic** (crysta) · derivatives come from the same formulas
accurate to numerical precision · extra work in the same evaluation

THREE WAYS TO GET DERIVATIVES

① **Numerical** (EasyDiffraction) · change, recalculate, subtract
finite differences · one extra full pattern per parameter, per step · approximate

② **Analytic** (FullProf) · derive formulas and write them by hand
every profile × every parameter · exact, but hundreds of formulas to maintain

③ **Automatic** (crysta) · derivatives come from the same formulas
accurate to numerical precision · extra work in the same evaluation

A number becomes a pair: (value, slope)

$$y = x^2 \text{ at } x = 3$$

$$(3, 1) \times (3, 1) = (9, 6)$$

DERIVATIVES CHANGE THE FIT

🧪 NCAF	📈 3977 points	⚙️ 627 peaks
📡 WISH 2+9	💻 VISA 6 cores	📏 linear scale
🚩 same start	✅ similar fit, $\chi^2 \sim 4.8$	⚙️ 48 free par.

DERIVATIVES CHANGE THE FIT

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

🖨️ VISA 6 cores

📏 linear scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

numerical (CrysPy + Imfit)

0 200 400 600 800



643 patterns

DERIVATIVES CHANGE THE FIT

🧪 NCAF

📈 3977 points

⚙️ 627 peaks

🔊 WISH 2+9

🖨️ VISA 6 cores

📏 linear scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

numerical (CrysPy + Imfit)

0 200 400 600 800

643 patterns

automatic (crysta)

37 equivalents

SPEED: A FULL REFINEMENT

🧪 NCAF

📡 WISH 2+9

🏳️ same start

📈 3977 points

💻 VISA 6 cores

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 627 peaks

📊 log scale

⚙️ 48 free par.

SPEED: A FULL REFINEMENT

🧪 NCAF

📡 WISH 2+9

🏠 same start

📈 3977 points

🖨️ VISA 6 cores

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 627 peaks

📊 log scale

⚙️ 48 free par.

CrysPy via EasyDiffraction (lmfit)

0.1 s

1 s

10 s

100 s

1000 s

468 s / 643 patterns

SPEED: A FULL REFINEMENT

🧪 NCAF

📈 3977 points

⚙️ 627 peaks

📡 WISH 2+9

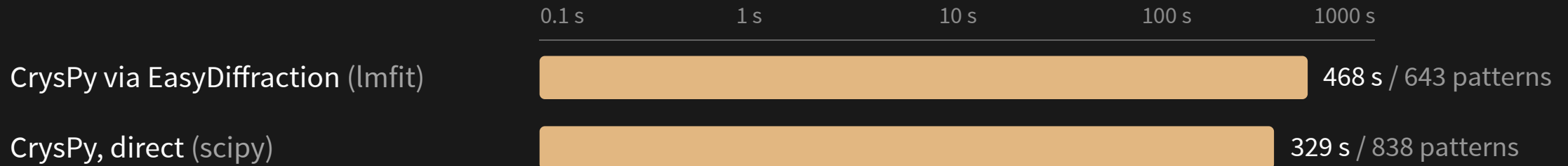
💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.



SPEED: A FULL REFINEMENT

🧪 NCAF

📡 WISH 2+9

🚩 same start

📈 3977 points

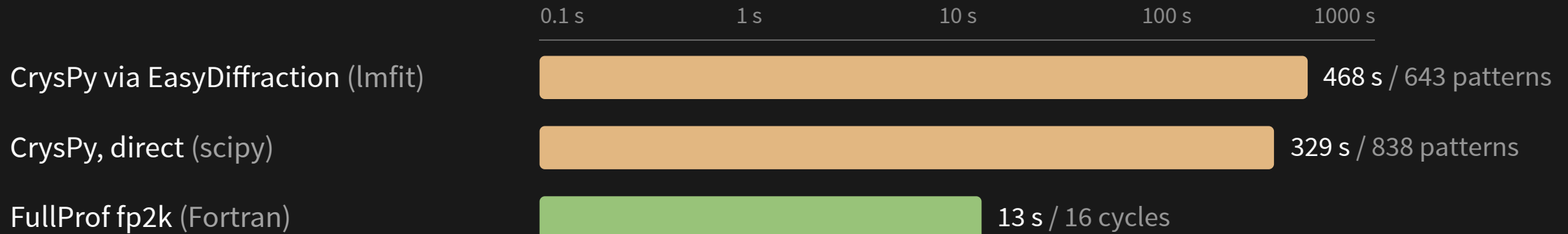
💻 VISA 6 cores

✅ similar fit, $\chi^2 \sim 4.8$

⬆️ 627 peaks

📊 log scale

⚙️ 48 free par.



SPEED: A FULL REFINEMENT

🧪 NCAF

📡 WISH 2+9

🚩 same start

📈 3977 points

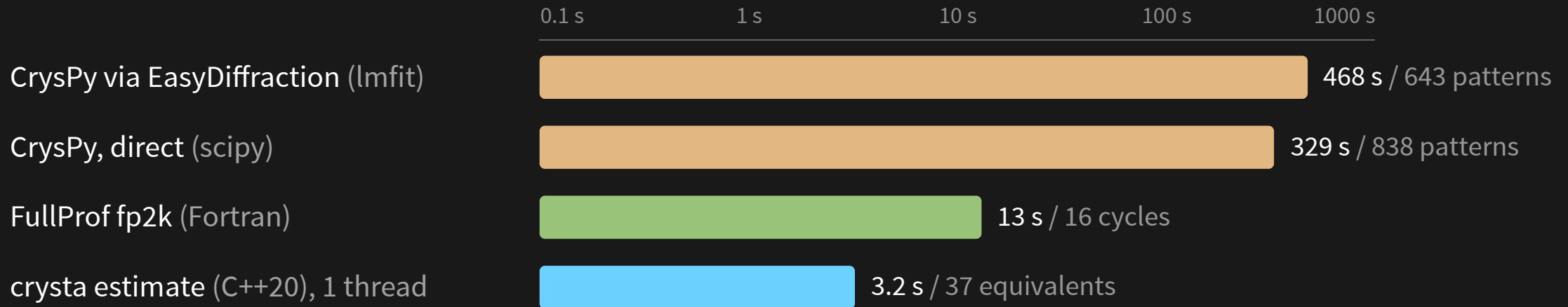
🔌 VISA 6 cores

✅ similar fit, $\chi^2 \sim 4.8$

⚠️ 627 peaks

📊 log scale

⚙️ 48 free par.



SPEED: A FULL REFINEMENT

🧪 NCAF

📡 WISH 2+9

🚩 same start

📈 3977 points

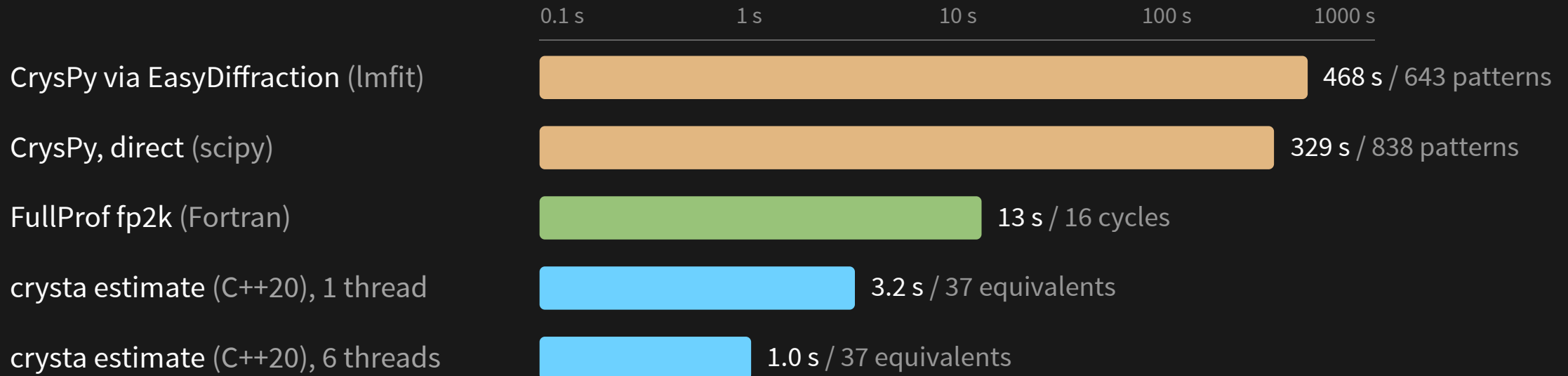
💻 VISA 6 cores

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 627 peaks

📊 log scale

⚙️ 48 free par.



AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📡 WISH 2+9

🚩 same start

📈 3977 points

💻 VISA 6 cores

👍 similar fit, $\chi^2 \sim 4.8$

⬆️ 627 peaks

📏 log scale

⚙️ 48 free par.

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

👍 similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	-0.000023	0.39

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📡 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	-0.000023	0.39
... 44 more ...				

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	-0.000023	0.39
... 44 more ...				
Ca1 B _{iso}	1.1183 ± 0.1268	1.1200 ± 0.1340	-0.0017	0.01

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	-0.000023	0.39
... 44 more ...				
Ca1 B _{iso}	1.1183 ± 0.1268	1.1200 ± 0.1340	-0.0017	0.01
F3 B _{iso}	1.8516 ± 0.1594	1.8520 ± 0.1690	-0.0004	0.00

AGREEMENT OF REFINED PARAMETERS

🧪 NCAF

📈 3977 points

⬆️ 627 peaks

📶 WISH 2+9

💻 VISA 6 cores

📊 log scale

🚩 same start

✅ similar fit, $\chi^2 \sim 4.8$

⚙️ 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	-23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	-0.000023	0.39
... 44 more ...				
Ca1 B _{iso}	1.1183 ± 0.1268	1.1200 ± 0.1340	-0.0017	0.01
F3 B _{iso}	1.8516 ± 0.1594	1.8520 ± 0.1690	-0.0004	0.00

Every one within 0.69σ · median 0.05σ

OUTLINE

💡 Motivation

🏗️ What was built

🕒 Speed and correctness

🔧 How it was built

THE RULES I STARTED WITH

THE RULES I STARTED WITH

- ⌵ Start from clear requirements, written by a human

THE RULES I STARTED WITH

- ☰ Start from clear requirements, written by a human
- 👤 Agree the plan before any code is written

THE RULES I STARTED WITH

- ☰ Start from **clear requirements**, written by a human
- 👤 Agree the plan **before any code is written**
- ✎ Write the tests **independently** of the code

THE RULES I STARTED WITH

- ☰ Start from **clear requirements**, written by a human
- 👤 Agree the plan **before any code is written**
- ✍ Write the tests **independently** of the code
- 🔄 Use **different models** to plan, implement and review

THE RULES I STARTED WITH

- ☰ Start from **clear requirements**, written by a human
- 👤 Agree the plan **before any code is written**
- ✎ Write the tests **independently** of the code
- 🔄 Use **different models** to plan, implement and review
- 🔄 Never trust **a single run** on anything complex

THE RULES I STARTED WITH

- ☰ Start from **clear requirements**, written by a human
- 👤 Agree the plan **before any code is written**
- ✎ Write the tests **independently** of the code
- 🔄 Use **different models** to plan, implement and review
- 🔄 Never trust **a single run** on anything complex
- 👤✓ **The human decides** what is correct and what is usable

FOUR AGENTS, ONE HUMAN

FOUR AGENTS, ONE HUMAN

 implement

Claude

FOUR AGENTS, ONE HUMAN

 implement

Claude

 write tests

GPT

FOUR AGENTS, ONE HUMAN

 implement

Claude

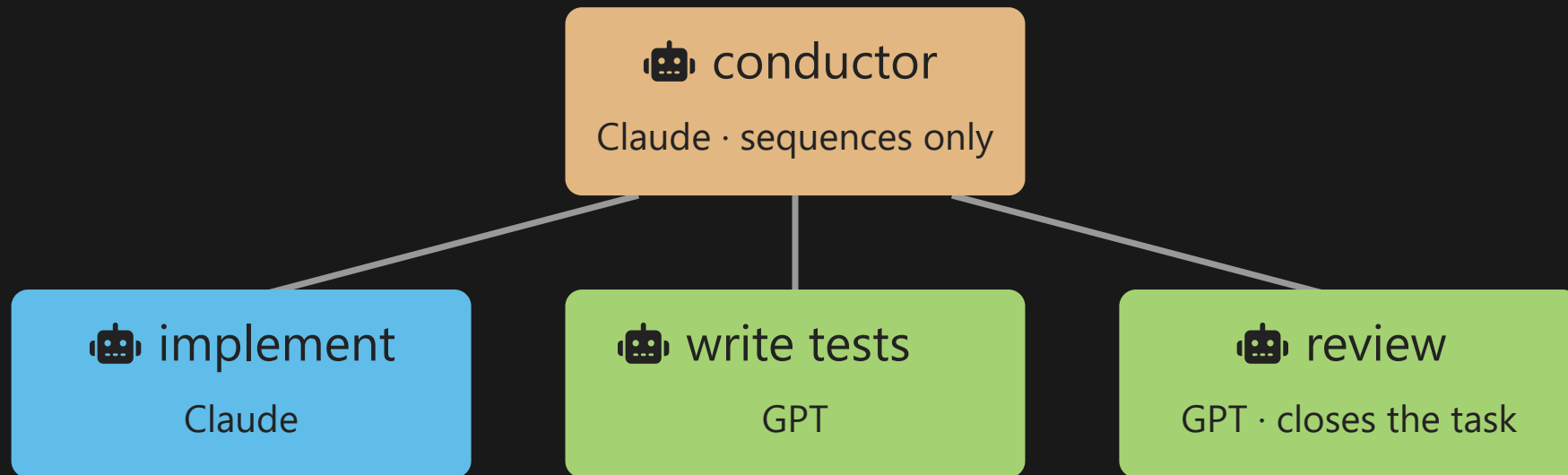
 write tests

GPT

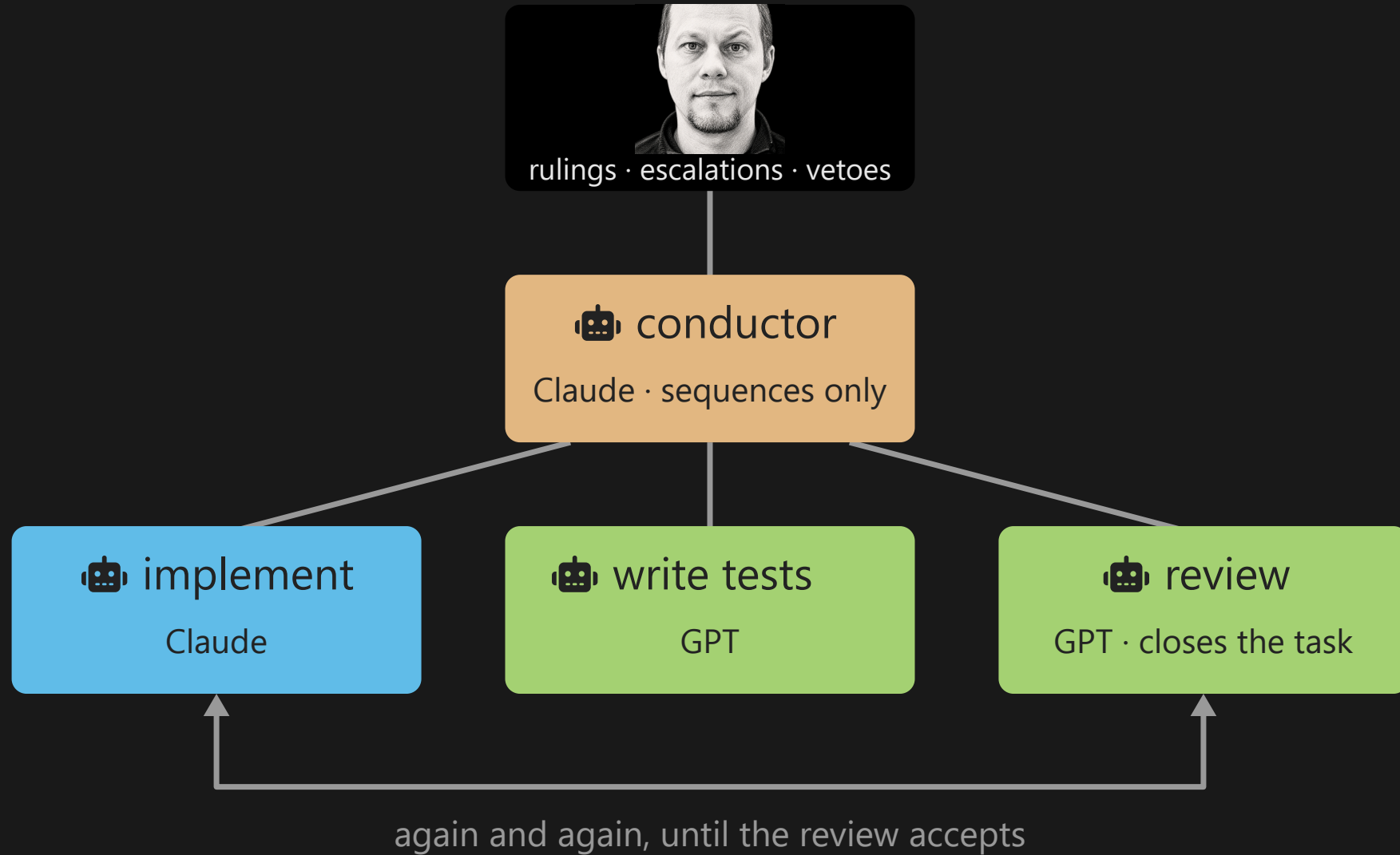
 review

GPT · closes the task

FOUR AGENTS, ONE HUMAN



FOUR AGENTS, ONE HUMAN



WHAT THE AGENT SAID

WHAT THE AGENT SAID

*"You're right, you've said it repeatedly and I **kept re-asking**."*

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

*"But I then **invented a merge gate** you never asked for."*

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

*"That's **your own ruling**, and **we** broke it."*

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

"That's your own ruling, and we broke it."

"I invented a coupling that doesn't exist and then designed a gate to police it."

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

*"That's your own ruling, and **we** broke it."*

"I invented a coupling that doesn't exist and then designed a gate to police it."

*"I ended that turn **saying 'Starting it' instead of starting it.**"*

RECURRING ERRORS NEED BETTER CHECKS

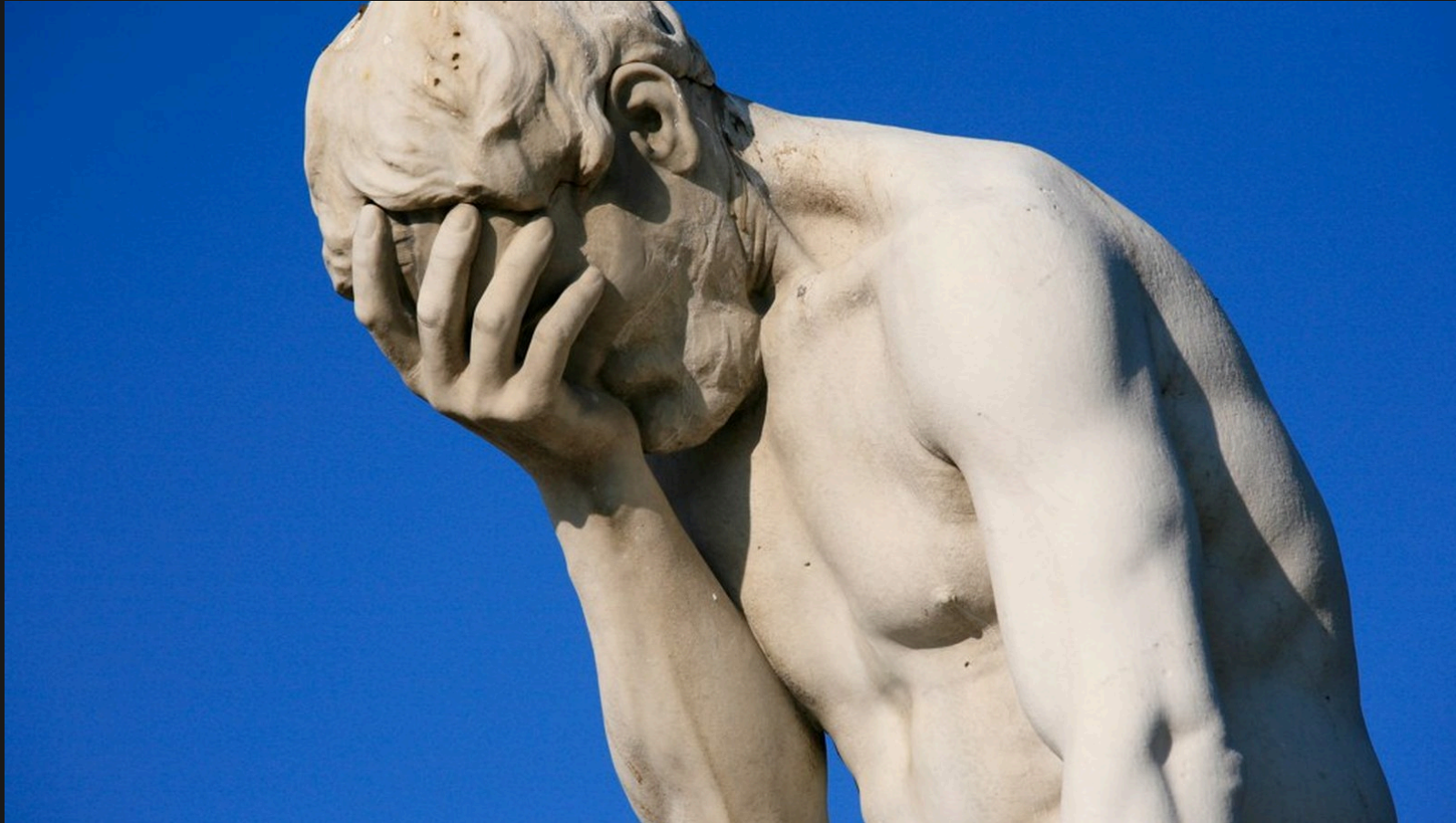
RECURRING ERRORS NEED BETTER CHECKS



Written down, and checked by a machine

... BUT IT HAPPENS AGAIN FROM TIME TO TIME

... BUT IT HAPPENS AGAIN FROM TIME TO TIME



Clear instructions do not prevent every error



**I did not write this code.
So I do not know where it is fragile.**

When you write something yourself, that knowledge is free — and it is a form of quality assurance.

Delegating the writing gives it up. Nothing replaces it unless you build the replacement on purpose.



**I did not write this code.
So I do not know where it is fragile.**

When you write something yourself, that knowledge is free — and it is a form of quality assurance.

Delegating the writing gives it up. Nothing replaces it unless you build the replacement on purpose.



A DIFFERENT ROUTE TO THE SAME QUESTION

AI-written CIF parser for CCTBX

Oleg Sobolev & Paul Adams · Lawrence Berkeley National Laboratory

MS-116 Teaching crystallographic computing · IUCr 2026, Calgary

HOW THE CODE IS CHECKED

HOW THE CODE IS CHECKED



one function or class at a time · C++ and Python

HOW THE CODE IS CHECKED

Unit tests

one function or class at a time · C++ and Python

Integration tests

components together, across the binding boundary

HOW THE CODE IS CHECKED

Unit tests

one function or class at a time · C++ and Python

Integration tests

components together, across the binding boundary

System tests

the command line and the project format, end to end

HOW THE CODE IS CHECKED

Unit tests

one function or class at a time · C++ and Python

Integration tests

components together, across the binding boundary

System tests

the command line and the project format, end to end

Calculation tests

test patterns against an [independent reference](#) · FullProf, CrysFML, CrysPy

HOW THE CODE IS CHECKED

Unit tests

one function or class at a time · C++ and Python

Integration tests

components together, across the binding boundary

System tests

the command line and the project format, end to end

Calculation tests

test patterns against an [independent reference](#) · FullProf, CrysFML, CrysPy

Fitting tests

test refinements against an [independent reference](#) · FullProf

TWO ERRORS FOUND BY TESTS

TWO ERRORS FOUND BY TESTS

THE TAILS VANISHED



The direct formula made the tails return **not a number**

a **very large** number times a **very small** one · the fix computes the same value without ever forming the large one

TWO ERRORS FOUND BY TESTS

THE TAILS VANISHED

 The direct formula made the tails return **not a number**

a **very large** number times a **very small** one · the fix computes the same value without ever forming the large one

WIDTH OR SPREAD

 It also mixed up two ways of measuring a peak's width

the width at half height is **2.35×** the Gaussian σ — the formula wanted σ

TWO ERRORS FOUND BY TESTS

THE TAILS VANISHED



The direct formula made the tails return **not a number**

a **very large** number times a **very small** one · the fix computes the same value without ever forming the large one

WIDTH OR SPREAD



It also mixed up two ways of measuring a peak's width

the width at half height is **2.35×** the Gaussian σ — the formula wanted σ

Both were found by tests written **before** the fix

CHECKS BEYOND THE TEST SUITE

CHECKS BEYOND THE TEST SUITE



C++ · every compiler warning is an error

static analysis and memory-safety checks on every change

CHECKS BEYOND THE TEST SUITE

 **C++** · every compiler warning is an error
static analysis and memory-safety checks on every change

 **Python** · strict style and type checks
exceptions need a reason and review

CHECKS BEYOND THE TEST SUITE

 **C++** · every compiler warning is an error
static analysis and memory-safety checks on every change

 **Python** · strict style and type checks
exceptions need a reason and review

 **Reproducibility** · matching numbers in tested configurations
checks vary thread count, vector width and build options

CHECKS BEYOND THE TEST SUITE

 **C++** · every compiler warning is an error
static analysis and memory-safety checks on every change

 **Python** · strict style and type checks
exceptions need a reason and review

 **Reproducibility** · matching numbers in tested configurations
checks vary thread count, vector width and build options

 **Every change** · CI runs automatically
Linux and macOS · checks depend on the change and event

WHAT YOU NEED BEFORE YOU START

WHAT YOU NEED BEFORE YOU START

- 🗄️ **Knowledge base** · the maths and physics, from literature and open code
structure factors, peak profiles, refinement · collected once, in one place

WHAT YOU NEED BEFORE YOU START

 **Knowledge base** · the maths and physics, from literature and open code
structure factors, peak profiles, refinement · collected once, in one place

 **Independent references** · other programs, run as black boxes
FullProf, CrysFML, cctbx ... compare outputs independently of the new code

WHAT YOU NEED BEFORE YOU START

-  **Knowledge base** · the maths and physics, from literature and open code
structure factors, peak profiles, refinement · collected once, in one place
-  **Independent references** · other programs, run as black boxes
FullProf, CrysFML, cctbx ... compare outputs independently of the new code
-  **Domain knowledge** · a human who understands the physics
judge whether the models and test cases make physical sense

WHAT YOU NEED BEFORE YOU START

-  **Knowledge base** · the maths and physics, from literature and open code
structure factors, peak profiles, refinement · collected once, in one place
-  **Independent references** · other programs, run as black boxes
FullProf, CrysFML, cctbx ... compare outputs independently of the new code
-  **Domain knowledge** · a human who understands the physics
judge whether the models and test cases make physical sense
-  **Software development skills** · to judge what the AI proposes
architecture, interfaces, and when a solution is over-engineered

WHAT YOU NEED BEFORE YOU START

-  **Knowledge base** · the maths and physics, from literature and open code
structure factors, peak profiles, refinement · collected once, in one place
-  **Independent references** · other programs, run as black boxes
FullProf, CrysFML, cctbx ... compare outputs independently of the new code
-  **Domain knowledge** · a human who understands the physics
judge whether the models and test cases make physical sense
-  **Software development skills** · to judge what the AI proposes
architecture, interfaces, and when a solution is over-engineered
-  **Time for review** · write the rules down, let a machine check them
a check that fails beats a cleverer prompt

CREDITS: LITERATURE

CREDITS: LITERATURE

Reference volumes	<i>International Tables for Crystallography</i> , Vol. A (symmetry) and Vol. C (tables) · ILL Neutron Data Booklet
Structure factors	Cromer & Mann 1968 · Waasmaier & Kirfel 1995 · Busing & Levy 1967
Peak profiles	Caglioti 1958 · David 1986 · Thompson, Cox & Hastings 1987 · Bérar & Baldinozzi 1993 · Finger, Cox & Jephcoat 1994
Time of flight	Von Dreele 1982 · Ikeda & Carpenter 1985
Corrections	Becker & Coppens 1974 · Hewat 1979 · Dollase 1986
Background	Ryan 1988 · Cobas 2006 · Baek 2015
Refinement	Rietveld 1969 · Marquardt 1963 · Moré 1978 · Toby 2006 · Coelho 2005, 2018
Total scattering	Proffen & Billinge 1999 · Jeong 1999 · Keen 2001 · Farrow 2007 · Egami & Billinge, <i>Underneath the Bragg Peaks</i>
Magnetism	Blume 1963 · Bertaut 1968 · Lisher & Forsyth 1971 · Brown 1979 · Wills 2000 · Rodríguez-Carvajal, lecture notes

87 papers and volumes · the goal: every equation links to its source

CREDITS: OPEN SOURCE SOFTWARE

CREDITS: OPEN SOURCE SOFTWARE

CrysPy

Iurii Kibalin · ESS

cctbx

Grosse-Kunstleve, Adams, Sauter et al. · LBNL

CCSL

P. J. Brown & J. C. Matthewman · ILL / CCDC

Dans_Diffraction

Dan Porter · Diamond Light Source

MAUD

Luca Lutterotti · University of Trento

CrysFML

Rodríguez-Carvajal, González-Platas, Katcho · ILL

GSAS-II

Toby & Von Dreele · Argonne

ObjCryst++ / Fox

Favre-Nicolin & Černý

periodictable

Paul Kienzle · NIST

Mantid

the Mantid collaboration · ISIS, ORNL, ILL, ESS

WHERE THIS LEAVES ME

WHERE THIS LEAVES ME

AI **helps** me build more

I can try ideas I did not have time for



WHERE THIS LEAVES ME

AI **helps** me build more

I can try ideas I did not have time for



It does **not** give me free time

checking the work still takes time



WHERE THIS LEAVES ME

AI **helps** me build more

I can try ideas I did not have time for



It does **not** give me free time

checking the work still takes time



AI can write code · people remain responsible for deciding **what counts as correct**

THANK YOU